**Project Work Phase-I Report on**

# "Prenatal Plus: Integrated Maternal, Fetal, and Pediatric Care "

*A Dissertation submitted in partial fulfilment of the requirement for the award of degree*

**MASTER OF COMPUTER APPLICATIONS**
OF
**VISVESVARAYA TECHNOLOGICAL UNIVERSITY**

By


**KISHOR RATHOD**

**1BY24MC032**

**III Semester**


**Under the Guidance of**

**Internal guide**


**Ashwitha K**
Asst. Professor
Department of MCA
BMSIT&M
Bengaluru-560119



**Department of Master of Computer Applications**
**(Accredited by NBA, New Delhi)**

**BMS Institute of Technology & Management**
**(Autonomous Institution under VTU)**
**Bengaluru – 560119**


**December-2025**

**Project Work Phase-I Report on**

# "Prenatal Plus: Integrated Maternal, Fetal, and Pediatric Care "

*A Dissertation submitted in partial fulfilment of the requirement for the award of degree*

**MASTER OF COMPUTER APPLICATIONS**
OF
**VISVESVARAYA TECHNOLOGICAL UNIVERSITY**

By


**BASAVANTH KUMAR PATTAR**

**1BY24MC013**

**III Semester**


**Under the Guidance of**

**Internal guide**


**Ashwitha K**
Professor
Department of MCA
BMSIT&M
Bengaluru-560119



**Department of Master of Computer Applications**
**(Accredited by NBA, New Delhi)**

**BMS Institute of Technology & Management**
**(Autonomous Institution under VTU)**
**Bengaluru – 560119**


**December-2025**

# BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

**(Autonomous Institution under VTU)**

**Bengaluru – 560119**

**DECEMBER-2025**

## CERTIFICATE

This is to certify that the dissertation of Project Work Phase-I titled "Prenatal Plus: Integrated Maternal, Fetal, and Pediatric Care" submitted in partial fulfilment of the requirements for the degree "Master of Computer Applications" by Visvesvaraya Technological University is based on an original study and is record of bonafide work carried out by Kishor Rathod bearing university registration number 1BY24MC032 during the period September 2025 to December 2025 under our supervision and guidance and that no part of the report has been submitted for the award of any other Degree/ Diploma/ Fellowship or similar title or prizes. The project report has been approved as it satisfies the academic requirements in respect of Project Work (Phase-I) of III Semester prescribed for the Master of Computer Applications Degree.

| **Signature of the Internal Guide** | **Signature of the HOD** | **Signature of the Principal** |
|---|---|---|
| **PROF. ASHWITHA. K** | **DR. M. SRIDEVI** | **DR. SANJAY H A** |
| ashwitha@bmsit.in | hod_mca@bmsit.in | principal@bmsit.in |
| Asst Professor | Asst. Professor & HOD | Principal |
| Department of MCA | Department of MCA | BMSIT&M |
| BMSIT&M | BMSIT&M | Bengaluru-560119 |
| Bengaluru-560119 | Bengaluru-560119 | |

**External Viva-Voice**

**Name of Examiners**

**1.** ---------- **Signature**

**2.** ---------- **Signature**

# BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

**(Autonomous Institution under VTU)**

**Bengaluru – 560119**

**DECEMBER-2025**

## CERTIFICATE

This is to certify that the dissertation of Project Work Phase-I titled "Prenatal Plus: Integrated Maternal, Fetal, and Pediatric Care" submitted in partial fulfilment of the requirements for the degree "Master of Computer Applications" by Visvesvaraya Technological University is based on an original study and is record of bonafide work carried out by Basavanth Kumar Pattar bearing university registration number 1BY24MC013 during the period September 2025 to December 2025 under our supervision and guidance and that no part of the report has been submitted for the award of any other Degree/ Diploma/ Fellowship or similar title or prizes. The project report has been approved as it satisfies the academic requirements in respect of Project Work (Phase-I) of III Semester prescribed for the Master of Computer Applications Degree.

| **Signature of the Internal Guide** | **Signature of the HOD** | **Signature of the Principal** |
|---|---|---|
| **ASHWITHA K** | **DR. M. SRIDEVI** | **DR. SANJAY H A** |
| ashwitha@bmsit.in | hod_mca@bmsit.in | principal@bmsit.in |
| Asst. Professor | Asst. Professor & HOD | Principal |
| Department of MCA | Department of MCA | BMSIT&M |
| BMSIT&M | BMSIT&M | Bengaluru-560119 |
| Bengaluru-560119 | Bengaluru-560119 | |

**External Viva-Voice**

**Name of Examiners**

**1.** ---------- **Signature**

**2.** ---------- **Signature**

# DECLARATION

I Kishor Rathod, III Semester student of MCA, BMS Institute of Technology & Management, bearing USN 1BY24MC032 hereby declared that project entitled "Prenatal Plus: Integrated Maternal, Fetal, and Pediatric Care" has been carried out by me under the supervision of internal guide, Prof. Ashwitha K and submitted in the partial fulfilment of the requirements for the award of Degree of Master of Computer Applications by the Visvesvaraya Technological University during the academic year 2025-26. This report has not been submitted to any other Organization/University for any award of degree or certificate.

Signature

Place: Bengaluru

Name: Kishor Rathod

Date:

USN: 1BY24MC032

# DECLARATION

I Basavanth Kumar Pattar, III Semester student of MCA, BMS Institute of Technology & Management, bearing USN 1BY24MC013 hereby declared that project entitled "Prenatal Plus: Integrated Maternal, Fetal, and Pediatric Care" has been carried out by me under the supervision of internal guide, Prof. Ashwitha K and submitted in the partial fulfilment of the requirements for the award of Degree of Master of Computer Applications by the Visvesvaraya Technological University during the academic year 2025-26. This report has not been submitted to any other Organization/University for any award of degree or certificate.

Signature

Place: Bengaluru

Name: Basavanth Kumar Pattar

Date:

USN: 1BY24MC013

# ACKNOWLEDGEMENT

I would like to take this opportunity to express my heartfelt gratitude to everyone who contributed to the successful completion of my project. I would especially like to acknowledge a few individuals to whom I owe sincere thanks and appreciation. First and foremost, I extend my sincere gratitude to the BMSIT Management for providing excellent infrastructure and educational support, which have greatly contributed to shaping my career.

I would also like to express my deep appreciation to our Principal, Dr. Sanjay H.A., for his kind support and guidance throughout the course of this project.

My heartfelt thanks go to Dr. M. Sridevi, Head of the Department, for her invaluable insights and support in making this project possible.

I am profoundly grateful to my internal guide, Prof. Ashwitha K for his unwavering support, encouragement, and valuable inputs throughout the entire process of completing this project.

I also extend my thanks to all the professors and non-teaching staff members who have contributed their help and support, directly or indirectly, in completing this project. Finally, I would like to express my deepest gratitude to my parents and friends for their moral support and encouragement, which helped me in accomplishing this project.

**Kishor Rathod**

**1BY24MC032**

# ACKNOWLEDGEMENT

I would like to take this opportunity to express my heartfelt gratitude to everyone who contributed to the successful completion of my project. I would especially like to acknowledge a few individuals to whom I owe sincere thanks and appreciation. First and foremost, I extend my sincere gratitude to the BMSIT Management for providing excellent infrastructure and educational support, which have greatly contributed to shaping my career.

I would also like to express my deep appreciation to our Principal, Dr. Sanjay H.A., for his kind support and guidance throughout the course of this project.

My heartfelt thanks go to Dr. M. Sridevi, Head of the Department, for her invaluable insights and support in making this project possible.

I am profoundly grateful to my internal guide, Prof. Ashwitha K for his unwavering support, encouragement, and valuable inputs throughout the entire process of completing this project.

I also extend my thanks to all the professors and non-teaching staff members who have contributed their help and support, directly or indirectly, in completing this project. Finally, I would like to express my deepest gratitude to my parents and friends for their moral support and encouragement, which helped me in accomplishing this project.

**Basavanth Kumar Pattar**

**1BY24MC013**

# BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

**(Autonomous Institution under VTU)**

**Bengaluru – 560119**

## Department of MCA

**(Accredited by NBA, New Delhi)**

## VISION

To emerge as a leading department in computer applications, producing skilled professionals equipped to deliver sustainable solutions.

## MISSION

Facilitate effective learning environment through quality education, industry interaction with orientation towards research, critical thinking and entrepreneurial skills.

## Programme Educational Objectives (PEOs)

**PEO 1:** Excel in IT career by developing sustainable solutions that drive industry growth and societal progress

**PEO 2:** Adapt themselves to evolving domain requirements

**PEO 3:** Exhibit leadership skills and progress in their chosen career path

# BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

**(Autonomous Institution under VTU)**

**Bengaluru – 560119**

## Department of MCA

**(Accredited by NBA, New Delhi)**

## Programme Outcomes (POs)

**PO 1:** Apply knowledge of mathematics, programming logic and coding fundamentals for solution architecture and problem solving.

**PO 2:** Identify, review, formulate and analyse problems for primarily focusing on customer requirements using critical thinking frameworks.

**PO 3:** Design, develop and investigate problems with an innovative approach for solutions incorporating ESG/SDG goals.

**PO 4:** Select, adapt and apply modern computational tools such as development of algorithms with an understanding of the limitations including human biases.

**PO 5:** Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups using methodologies such as agile.

**PO 6:** Use the principles of project management such as scheduling, work breakdown structure and be conversant with the principles of Finance for profitable project management.

**PO 7:** Commit to professional ethics in managing software projects with financial aspects. Learn to use new technologies for cyber security and insulate customers from malware.

**PO 8:** Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

# BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

**(Autonomous Institution under VTU)**

**Bengaluru – 560119**

## Department of MCA

**(Accredited by NBA, New Delhi)**

## CO – PO Mapping

| CO/PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 |
|---|---|---|---|---|---|---|---|---|
| Synopsis | 3 | 3 | 3 | 1 | 2 | 2 | 1 | 2 |
| Interim Presentation | 3 | 3 | 3 | 1 | 2 | 1 | 1 | 2 |
| Interim Demonstration | 3 | 3 | 3 | 3 | 2 | 2 | 2 | 2 |
| Final Demonstration | 3 | 2 | 3 | 3 | 2 | 3 | 2 | 2 |
| Report Evaluation | 3 | 2 | 2 | 2 | 3 | 2 | 3 | 3 |

# ABSTRACT

Prenatal healthcare plays a vital role in ensuring the physical and emotional well-being of both the expectant mother and the developing fetus. Regular medical checkups, continuous monitoring of health conditions, proper nutrition, and timely medical guidance are essential components of effective prenatal care. When these practices are followed consistently, many pregnancy-related complications can be prevented or identified at an early stage. Despite medical advancements, a large number of expectant mothers still depend on manual health records, periodic hospital visits, and verbal instructions from healthcare professionals. Such practices often result in missed appointments, incomplete medical history maintenance, and limited awareness of prenatal health practices. These challenges highlight the need for a structured and reliable digital system to support prenatal care activities.

PrenatalPlus is developed as a digital prenatal healthcare support system that aims to simplify and improve the management of pregnancy-related information. The application provides a centralized platform where pregnant women can securely store personal and health-related data, monitor different stages of pregnancy, and receive reminders for scheduled medical checkups. The system also offers access to reliable prenatal health information, helping users make informed decisions regarding their health. By organizing data in a digital format, PrenatalPlus reduces dependence on paper-based records, minimizes the risk of data loss, and improves continuity of care. The user-friendly interface ensures ease of use, making the system accessible even to users with limited technical knowledge.

This Phase-I report focuses on analyzing the prenatal healthcare domain and identifying limitations in existing prenatal care systems. It includes a detailed study of current practices, feasibility analysis, and identification of functional and non-functional requirements for the proposed system. The report also presents the architectural design and system planning of PrenatalPlus. The work carried out in this phase mainly concentrates on requirement analysis and system design, which are essential for successful development. The outcomes of Phase-I provide a strong foundation for implementation, testing, and further enhancement of the system in Phase-II.



**Keywords:** Prenatal Healthcare, Maternal Health, Digital Health System, Pregnancy Monitoring, Web Application

# TABLE OF CONTENTS


| | Page No. |
|---|---|
| **1. INTRODUCTION** | **1** |
| 1.1.Introduction | 1 |
| 1.2.Need for the System | 2 |
| 1.3.Project Description | 2 |
| 1.4.Objectives of the Project | 3 |
| 1.5.Scope of the Project | 3 |
| **2. LITERATURE SURVEY** | **4** |
| 2.1.Existing and Proposed System | 4 |
| 2.2.Feasibility study | 6 |
| 2.2.1. Technical feasibility study | 6 |
| 2.2.2. Operational feasibility study | 7 |
| 2.2.3. Economic feasibility study | 7 |
| 2.3.Tools and Technologies used | 7 |
| 2.3.1. React | 7 |
| 2.3.2. Node.js and Express.js | 8 |
| 2.3.3. MongoDB | 8 |
| 2.3.4. Python | 8 |
| 2.3.5. Pandas and Matplotlib | 8 |
| 2.4.Hardware and Software requirements | 8 |
| **3. SOFTWARE REQUIREMENT SPECIFICATION** | **10** |
| 3.1.Users | 10 |
| 3.1.1. Scope and Objective | 10 |
| 3.1.2. Assumptions and Dependencies | 11 |
| 3.2.Functional requirements | 12 |
| 3.3.Non Functional requirements | 13 |
| **4. SYSTEM DESIGN** | **15** |
| 4.1.System Architecture | 15 |
| 4.2.Context Diagram Level 0 | 16 |
| 4.3.Context Diagram for Users | 17 |
| 4.4.Context Diagram for Admin | 18 |

**5. DETAILED DESIGN** **19**
5.1.Dataflow diagram 19
5.2.Activity diagram 20
5.3.Use Case diagram 22
5.4.Sequence diagram 23
**6. IMPLEMENTATION** **25**
6.1.Snippet code 25
6.2.Screenshots 35
**7. SOFTWARE TESTING** **40**
7.1.Unit Testing 40
7.2.Automation testing 41
7.3.Test Cases 41
**8. CONCLUSION** **43**
**9. SCOPE FOR PHASE-II** **45**
**10. BIBLIOGRAPHY** **47**
**11. APPENDIX A: USER MANUAL** **48**
**12. APPENDIX B: SDG COMPLIANCE** **51**
**13. APPENDIX C: WORK BREAKDOWN STRUCTURE** **53**
**14. APPENDIX D: PLAGIARISM REPORT** **57**

# LIST OF FIGURES


| Particulars | Page No. |
|---|---|
| 4.1 System Architecture | 15 |
| 4.2 Context Diagram Level 0 | 16 |
| 4.3 Context Diagram for Users | 17 |
| 4.4 Context Diagram for Admin | 18 |
| 5.1 Data Flow Diagram | 19 |
| 5.2 Activity Diagram | 20 |
| 5.3 Use Case Diagram | 22 |
| 5.4 Sequence Diagram | 23 |
| 6.1 Project Structure | 25 |
| 6.2 Login Page | 35 |
| 6.3 User Registration Page | 36 |
| 6.4 User Dashboard | 36 |
| 6.5 Pregnancy Profile Management Page | 37 |
| 6.6 Appointment Scheduling Page | 37 |
| 6.7 Reminder Notification Display | 38 |
| 6.8 Admin login page | 38 |
| 6.9 Admin Dashboard | 39 |
| 6.10 User Management Module | 39 |
| 13.1 SDLC Work Breakdown Structure | 53 |
| 13.2 Project Phase Timeline | 55 |

## LIST OF TABLES


| Particulars | Page No. |
|---|---|
| 2.1 Literature Survey | 5 |
| 7.1 Test Cases | 41 |
| 9.1 Phase -II Mobile App Development Timeline | 46 |
| 12.1 Mapping of SDG Goals | 51 |
| 12.2 Inter/Multidisciplinary Aspects Involved in the Project | 51 |
| 13.1 Project Timeline | 56 |

# 1. INTRODUCTION

## 1.1 Introduction

Healthcare is one of the most important sectors where accuracy, accessibility, and timely decision-making are essential. In recent years, the integration of information technology into healthcare has significantly improved the way medical services are delivered and managed. Digital healthcare systems help in storing patient data, improving communication, and supporting informed medical decisions. Among the various healthcare domains, maternal healthcare requires continuous attention as it directly affects the health and safety of both the mother and the unborn child.

Prenatal care is a critical phase of maternal healthcare that involves regular medical checkups, monitoring of health parameters, proper nutrition, and timely medical guidance throughout pregnancy. Effective prenatal care helps in early detection of complications such as anemia, gestational diabetes, and high blood pressure, thereby reducing health risks during pregnancy and childbirth. However, despite advancements in medical science, many expectant mothers still face challenges in accessing continuous and organized prenatal care.

In several healthcare environments, prenatal care management continues to rely on traditional methods such as handwritten medical records, physical reports, and periodic hospital visits. These approaches often lead to incomplete record maintenance, difficulty in tracking medical history, and missed medical appointments. Lack of timely reminders and limited access to reliable prenatal health information further increase the risk of poor maternal health outcomes. These limitations highlight the need for a digital solution that supports continuous monitoring and structured healthcare management during pregnancy.

The rapid growth of internet connectivity and digital devices has created opportunities for developing web-based healthcare applications that improve patient engagement and healthcare accessibility. Digital platforms allow users to access healthcare services and information at any time, reducing dependency on physical visits for routine guidance. For expectant mothers, such systems can provide continuous support, reminders, and access to prenatal health information, which is essential for maintaining a healthy pregnancy.

## 1.2 Need for the System

Maternal healthcare involves multiple factors such as regular monitoring, lifestyle management, and adherence to medical advice. Without a proper system to track and manage these aspects, pregnant women may overlook important healthcare routines. Many expectant mothers find it difficult to remember scheduled checkups, follow dietary guidelines, and monitor pregnancy-related changes consistently. This lack of structured support can negatively impact maternal and fetal health.

Another major challenge in prenatal care is the absence of centralized storage for medical records. Health information is often scattered across multiple documents and healthcare centers, making it difficult to access a complete medical history when required. In emergency situations, the lack of organized medical data can delay treatment and decision-making. A centralized digital system helps in maintaining accurate and easily accessible health records, ensuring better continuity of care.

Digital prenatal healthcare systems also help in improving awareness among expectant mothers. Access to reliable and structured prenatal information enables users to understand pregnancy-related changes and follow recommended healthcare practices. By providing reminders and guidance, such systems encourage proactive healthcare behavior, which plays a key role in improving pregnancy outcomes.

## 1.3 Project Description

PrenatalPlus is a web-based digital prenatal healthcare support system developed to assist expectant mothers throughout their pregnancy journey. The system provides a centralized platform where users can securely store and manage pregnancy-related information. It supports features such as pregnancy stage tracking, appointment reminders, and access to prenatal healthcare guidance.

The application is designed to be simple and user-friendly, allowing users with basic digital knowledge to operate the system easily. PrenatalPlus reduces dependency on paper-based medical records and ensures accurate organization of health data. The system also supports administrators in managing user information and maintaining system integrity.

By integrating healthcare information and reminders into a single platform, PrenatalPlus improves communication, reduces manual effort, and enhances overall prenatal care management. The system aims to support both expectant mothers and healthcare providers by promoting organized and continuous healthcare practices.

## 1.4 Objectives of the Project

The primary objective of PrenatalPlus is to design and develop a digital system that enhances prenatal healthcare management. The project aims to provide structured support for expectant mothers by offering timely reminders and access to reliable healthcare information. Another objective is to improve awareness regarding prenatal health practices and encourage proactive healthcare behavior.

The system also aims to ensure secure storage and management of sensitive health data. By providing a centralized digital platform, PrenatalPlus seeks to improve accessibility, reduce manual record-keeping, and support better healthcare decision-making during pregnancy.

## 1.5 Scope of the Project

The scope of PrenatalPlus includes digital management of prenatal healthcare information and support services. The system focuses on pregnancy tracking, appointment reminders, and healthcare guidance. It is designed to be scalable, allowing additional features such as advanced analytics, teleconsultation, and mobile application support to be integrated in future phases.

The current phase of the project concentrates on requirement analysis, system design, and architectural planning. Implementation, testing, and feature enhancement will be carried out in subsequent phases.

# 2. LITERATURE SURVEY

## 2.1 Existing and Proposed System

**Existing System**

The existing prenatal healthcare system in many regions, especially in developing countries, is largely fragmented and limited in scope. Traditional prenatal care practices mainly focus on periodic hospital visits, basic medical checkups, and manual record keeping. Most healthcare facilities rely on paper-based medical files to record patient history, diagnostic reports, and prescriptions. These records are usually stored locally and accessed only during physical consultations, which restricts continuous monitoring of the mother's health throughout pregnancy.

Although medical guidelines and prenatal care protocols exist, their implementation is often inconsistent. Expectant mothers are expected to remember appointment schedules, follow dietary and lifestyle instructions, and monitor symptoms without structured support. In rural and under-resourced areas, lack of awareness, limited access to healthcare facilities, and inadequate follow-up mechanisms further reduce the effectiveness of prenatal care. Sociocultural factors, misinformation, and lack of family or community support also act as barriers in seeking timely medical attention.

While some hospitals use digital hospital management systems, these systems primarily handle administrative tasks such as billing and registration. They rarely address patient-centric prenatal needs such as pregnancy stage tracking, automated reminders, and continuous health awareness. As a result, prenatal healthcare remains disjointed, reactive, and heavily dependent on manual processes.

**Proposed System**

The proposed system, PrenatalPlus, adopts a comprehensive and patient-centric approach to prenatal healthcare management. Unlike the existing system, PrenatalPlus focuses on providing continuous digital support to expectant mothers throughout different stages of pregnancy. The system is designed to integrate medical information, pregnancy tracking, appointment reminders, and health guidance into a single centralized platform.

PrenatalPlus emphasizes a holistic approach by addressing not only medical aspects but also awareness and routine healthcare practices. The system enables users to store pregnancy-related data securely, monitor progress, and receive timely notifications for checkups and essential prenatal activities. By digitizing prenatal care processes, the proposed system reduces dependency on paper records and minimizes the chances of missed appointments and incomplete medical history.

In addition, PrenatalPlus aims to improve accessibility and awareness by presenting healthcare information in a simple and user-friendly manner. The system supports proactive healthcare behavior and encourages regular follow-ups, thereby contributing to better maternal and fetal health outcomes. Overall, the proposed system serves as a reliable digital support tool for improving the quality and continuity of prenatal healthcare.

**Table 2.1: Literature Survey Table**

| Year | Author(s) | Title | Methodology | Key Findings |
|---|---|---|---|---|
| 2021 | R. Patel, S. Mehra | Digital Prenatal Care Monitoring System | Web-based healthcare platform with pregnancy tracking and reminders | Improved appointment adherence and prenatal health awareness among expectant mothers |
| 2020 | J. Smith, A. Brown | Mobile Health Application for Maternal Care | Android-based mHealth application with reminders and health tips | Increased user engagement and timely prenatal checkups |
| 2019 | L. Johnson, M. Clark | Prenatal Health Risks and Early Detection Techniques | Clinical studies and survey-based analysis | Identified the importance of early detection of pregnancy-related complications |
| 2022 | K. Rao, P. Nair | Smart Maternal Health Monitoring Using IoT | Sensor-based health monitoring with real-time data analysis | Demonstrated effective monitoring of vital parameters during pregnancy |

| | | | | |
|---|---|---|---|---|
| 2020 | Y. Chen, H. Wang | Emotion-Aware Mental Health Support for Pregnant Women | Sentiment analysis and chatbot-based interaction | Provided personalized mental health guidance during pregnancy |
| 2021 | N. Verma, S. Gupta | A Digital Pregnancy Journal for Expectant Mothers | Web-based journal system with health and mood tracking | Helped mothers track physical and emotional changes throughout pregnancy |

## 2.2 Feasibility Study

A feasibility study evaluates whether the proposed project can be successfully implemented within the constraints of available technology, operational conditions, and budget. This section analyzes the feasibility of developing and deploying the PrenatalPlus system by examining technical, operational, and economic aspects. The objective is to determine whether the project is practical, sustainable, and suitable for real-world application.

### 2.2.1 Technical Feasibility Study

The PrenatalPlus project is technically feasible as it utilizes widely available and stable web development technologies. The system is developed using standard frontend and backend frameworks that are well-supported by the developer community. No specialized hardware is required for development or deployment, and the application can run on commonly available computing devices.

The technologies used for frontend development, backend processing, and database management are compatible with modern web environments. Internet connectivity and basic computing infrastructure are sufficient for system operation. Since the project does not involve complex hardware integration or real-time medical devices in Phase-I, the technical risk associated with the project is minimal. Therefore, the proposed system is considered technically viable.

### 2.2.2 Operational Feasibility Study

Operational feasibility examines whether the system can be effectively used and managed in the intended environment. PrenatalPlus is designed to be user-friendly and intuitive, requiring minimal training for users. Expectant mothers can easily interact with the system to access information, receive reminders, and manage their pregnancy-related data.

From an administrative perspective, the system can be operated with minimal resources. The development and maintenance processes are straightforward and can be handled by a small development team. Since the application is web-based, it can be accessed from different locations without complex installation procedures. These factors make PrenatalPlus operationally feasible in both academic and practical healthcare settings.

### 2.2.3 Economic Feasibility Study

The proposed project is economically feasible as it involves low development and operational costs. PrenatalPlus uses open-source technologies and free development tools, which significantly reduce expenses. There is no requirement for costly hardware, proprietary software licenses, or specialized infrastructure.

The primary costs associated with the project include basic development resources and optional hosting services. Compared to traditional healthcare systems or large-scale clinical solutions, PrenatalPlus offers a cost-effective digital alternative for prenatal care management. Hence, the project is financially viable and suitable for academic as well as real-world deployment.

## 2.3 Tools and Technologies Used

### 2.3.1 React (Frontend)

React is a popular JavaScript library used for building user interfaces. In the PrenatalPlus project, React is used to develop the frontend components of the application. It enables the creation of a responsive and interactive user interface, improving user experience. The component-based architecture of React also makes the system scalable and easier to maintain.

### 2.3.2 Node.js and Express.js (Backend)

Node.js is a JavaScript runtime environment used for server-side development. Express.js, a framework built on Node.js, is used to manage API routes and handle client-server communication. This backend setup provides efficient request handling, smooth data processing, and reliable interaction with the database.

### 2.3.3 MongoDB (Database)

MongoDB is a NoSQL, document-oriented database used to store user data and pregnancy-related information. Its flexible schema allows efficient storage of structured and semi-structured data. MongoDB is scalable and well-suited for handling healthcare data securely and efficiently.

### 2.3.4 Python (Data Processing)

Python is used for data processing and backend utilities in the PrenatalPlus system. Its simplicity and wide range of libraries support data handling, validation, and future analytical enhancements.

### 2.3.5 Pandas and Matplotlib

Pandas is used for managing and analyzing datasets, while Matplotlib is used for visualizing data trends and reports. These tools support data analysis and performance evaluation during development and testing phases.

## 2.4 Hardware and Software Requirements

### 2.4.1 Hardware Requirements

- Development System: Computer with a minimum of 8 GB RAM (16 GB recommended).
- Processor: Intel i3/i5 or AMD Ryzen 5 or higher.

### 2.4.2 Software Requirements

- **Operating System:**

Windows 10/11, Ubuntu Linux, or macOS for development and deployment of the PrenatalPlus web application.

- **Programming Language:**
  JavaScript (ES6 or above) for both frontend and backend development.
- **Development Environment:**
  - Visual Studio Code: For writing, debugging, and managing frontend and backend source code.
  - Node.js Runtime Environment: For executing server-side JavaScript code and managing backend services.
- **Libraries and Frameworks:**
  - React.js: For developing a dynamic and responsive user interface.
  - Node.js: For handling server-side logic and request processing.
  - Express.js: For creating RESTful APIs and managing routing.
  - MongoDB: For secure and scalable storage of user and pregnancy-related data.
  - Mongoose: For object data modeling and database interaction.
- **Web Framework:**
  Express.js: For developing backend services and connecting the frontend with the database.
- **Web Browser:**
  Google Chrome, Mozilla Firefox, or Microsoft Edge for accessing and testing the PrenatalPlus web application.

# 3. SOFTWARE REQUIREMENT SPECIFICATION

## 3.1 Users

The system is intended for the following categories of users:

- **Expectant Mothers:**
  Pregnant women who wish to digitally manage their prenatal care, track pregnancy progress, receive reminders for medical checkups, and access reliable prenatal health information.
- **Healthcare Support Staff (Optional):**
  Medical staff or healthcare assistants who may assist users in understanding prenatal guidelines and ensuring regular follow-ups through the system.
- **Academic and Research Users:**
  Students and researchers who analyze digital healthcare systems for academic learning, project demonstrations, and research studies related to maternal healthcare management.
- **System Administrators:**
  Users responsible for managing system content, maintaining data integrity, and ensuring smooth operation of the PrenatalPlus application.
- **Public Health Awareness Groups:**
  Organizations or individuals who study the effectiveness of digital tools in improving prenatal awareness and healthcare accessibility.

### 3.1.1 Scope and Objective

**Scope**

PrenatalPlus is a user-friendly web-based prenatal healthcare support system designed to assist expectant mothers throughout pregnancy. The scope of the application is limited to the following key functionalities:

- Pregnancy profile creation and management to store essential pregnancy details.
- Appointment scheduling and reminder notifications for regular prenatal checkups.

- Access to reliable prenatal health information related to nutrition, lifestyle, and medical care.
- Secure storage and retrieval of pregnancy-related data.
- User authentication to ensure privacy and personalized access.

The application is developed as a web-based system using React.js for the frontend and Node.js with MongoDB for backend and database management.

**Objective**

The objectives of the PrenatalPlus project are:

- To provide a digital platform that supports expectant mothers in managing prenatal healthcare activities efficiently.
- To reduce missed medical appointments by implementing automated reminder notifications.
- To improve awareness of prenatal healthcare practices through reliable and easy-to-understand information.
- To design a simple and intuitive user interface suitable for users with minimal technical knowledge.
- To ensure secure handling of sensitive healthcare data with proper authentication and data protection mechanisms.
- To enable scalability for future enhancements such as mobile access, analytics, and integration with healthcare services.

### 3.1.2 Assumptions and Dependencies

**Assumptions**

- Users have access to a smartphone, tablet, or computer with a stable internet connection.
- Users provide accurate and truthful information related to pregnancy details.
- Users possess basic knowledge of using web applications.
- The application is intended to support healthcare management and not replace professional medical advice.

- Each user account is accessed by a single individual to maintain data privacy.

**Dependencies**

- Continuous internet connectivity is required for communication between frontend and backend services.
- The backend server must be active to process user requests and manage data.
- MongoDB must be available for storing and retrieving user and pregnancy-related information.
- Proper communication between React frontend and Node.js backend must be maintained through APIs.
- The system depends on third-party libraries and frameworks functioning correctly.

## 3.2 Functional Requirements

**User Registration and Login**

- Users shall be able to register using basic details such as name, email address, and password.
- Users shall be able to log in securely and maintain authenticated sessions.

**Pregnancy Profile Management**

- Users shall be able to enter and update pregnancy-related details such as due date and trimester information.
- The system shall store pregnancy data securely in the database.

**Appointment and Reminder System**

- Users shall be able to view upcoming prenatal checkups.
- The system shall send reminders for scheduled medical appointments.

**Health Information Access**

- Users shall be able to access prenatal healthcare tips and guidelines.

- The system shall display information related to nutrition, lifestyle, and general pregnancy care.

**User Authentication**

- Users shall register, log in, and securely access personalized features.
- User session data shall be protected and unique to each user.

## 3.3 Non-Functional Requirements

**Usability**

The system interface must be simple, intuitive, and easy to use for all users, including those with limited technical knowledge. Navigation should be consistent across all modules.

**Performance**

The application should load within 3 seconds under normal network conditions. User actions such as form submission and page navigation should respond within acceptable time limits.

**Availability**

The system should be available 24×7 with minimal downtime. Maintenance activities should be scheduled during non-peak hours.

**Scalability**

The system must be capable of handling an increasing number of users and data records without degrading performance.

**Security**

User data must be stored securely with proper authentication mechanisms. Unauthorized access to sensitive healthcare data must be prevented.

**Maintainability**

The system code should be modular and well-organized to allow easy maintenance, updates, and future enhancements.

**Reliability**

The system should operate continuously with minimal failures and ensure consistent performance during normal usage.

# 4. SYSTEM DESIGN

## 4.1 System Architecture

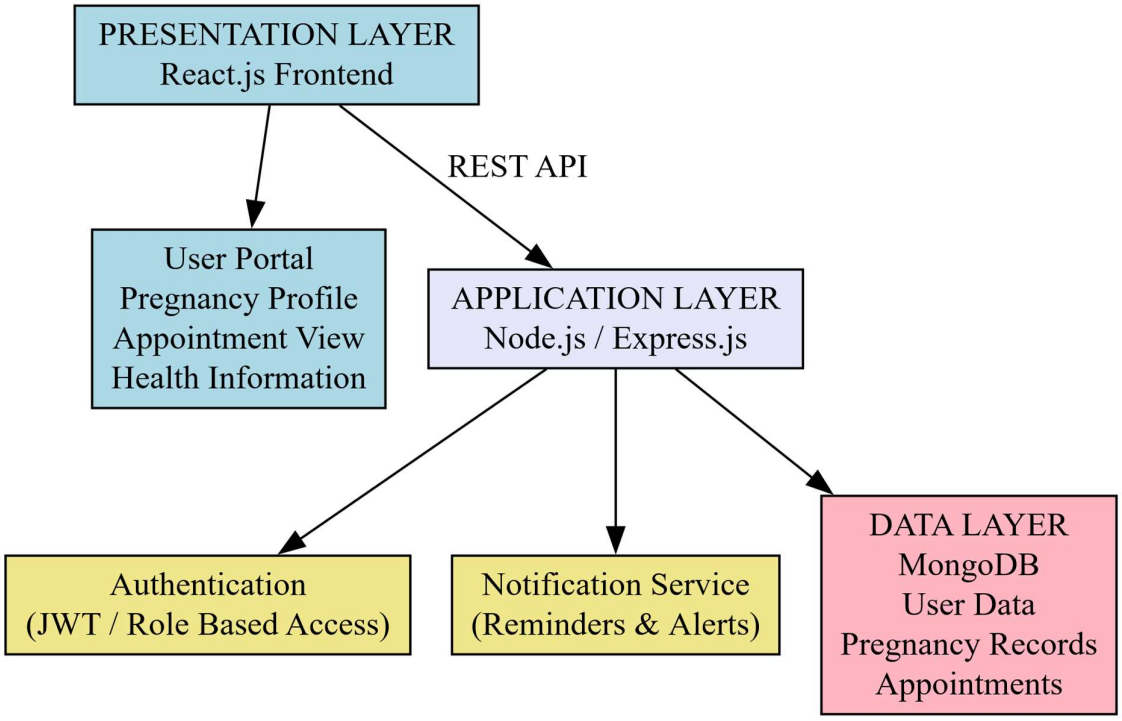


**Fig. 4.1 : System Architecture**

## Layered Design Pattern:

- **Presentation Layer (React.js SPA):**
  The presentation layer is implemented as a single-page application using React.js. It provides role-specific components for expectant mothers and system administrators. The user interface includes modules such as pregnancy profile management, appointment view, reminder notifications, and access to prenatal health information. React hooks are used for efficient state management, and responsive design principles ensure usability across devices.

- **Application Layer (Node.js/Express):**
  The application layer is developed using Node.js with the Express framework. It functions as a stateless RESTful backend that processes client requests through a

middleware pipeline including authentication, authorization, and controller logic. This layer manages pregnancy data, user sessions, reminder scheduling, and communication between the frontend and the database.

- **Data Layer (MongoDB):**
  The data layer uses MongoDB, a document-oriented NoSQL database, to store user profiles, pregnancy details, appointment schedules, and reminder history. Embedded documents are used to maintain pregnancy records per user. Indexing mechanisms are applied to frequently accessed fields to ensure efficient querying and fast dashboard loading.

## 4.2 Context Diagram Level 0

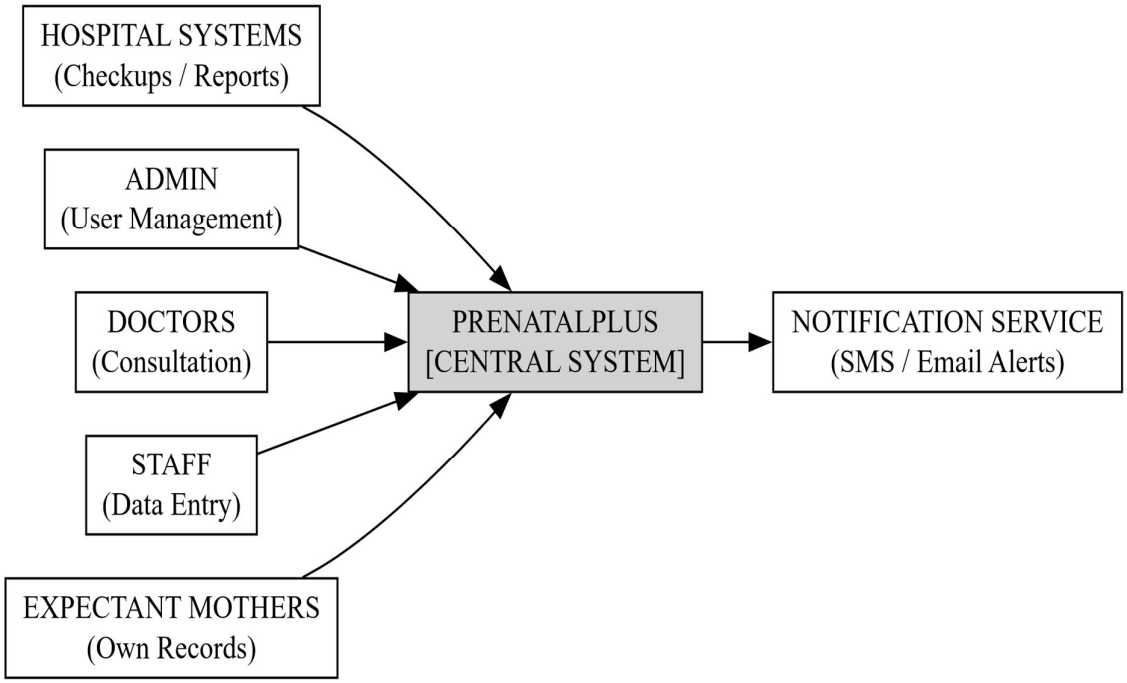


**Fig. 4.2 : Context Diagram Level 0**

The Level 0 Context Diagram represents PrenatalPlus as a single centralized system interacting with all major external entities involved in prenatal healthcare management. The PrenatalPlus system receives, processes, and stores pregnancy-related data entered by expectant mothers. It also delivers reminders, pregnancy progress updates, and healthcare information through the application interface.

Administrators interact with the system to manage users and maintain system data. Expectant mothers use the platform to enter pregnancy details, view scheduled checkups, and access prenatal guidance. The system also interacts with notification services to deliver reminder alerts. Through these interactions, PrenatalPlus acts as a central hub that connects users, data storage, and healthcare information delivery in a unified digital environment.

## 4.3 Context Diagram for Users

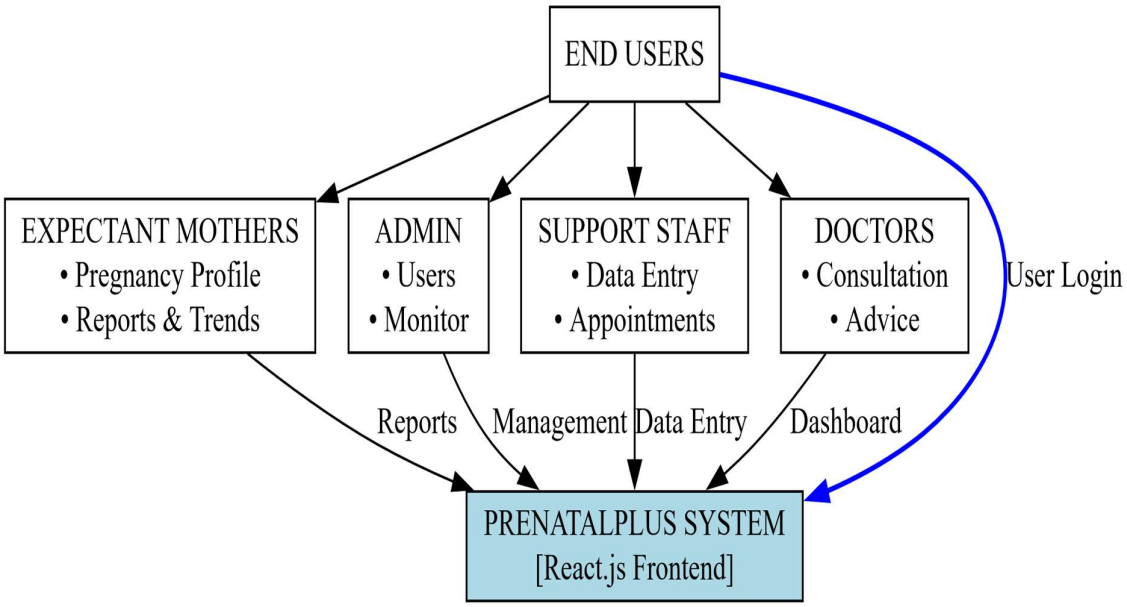


**Fig. 4.3 : Context Diagram for Users**

The user context diagram illustrates how different users interact with the PrenatalPlus system through a centralized React.js-based web application. Users primarily include expectant mothers and system administrators. All users are required to authenticate securely before accessing role-specific functionalities.

Expectant mothers interact with the system to manage their pregnancy profiles, view reminders for medical checkups, and access prenatal healthcare information. The system processes user input, updates pregnancy progress, and displays relevant information in a personalized manner. User requests and system responses occur through two-way communication, ensuring timely updates and accurate data presentation.

The PrenatalPlus system serves as the central processing unit for all user interactions, maintaining data consistency and delivering customized outputs based on user roles.

## 4.4 Context Diagram for Admin

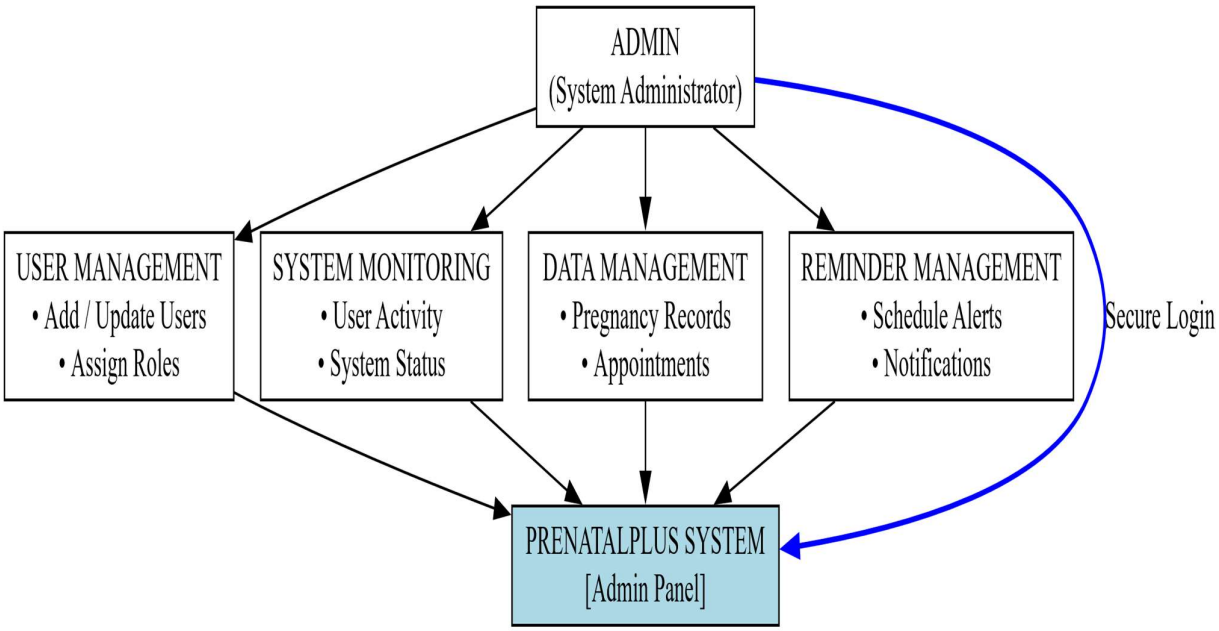


**Fig. 4.4 : Context Diagram for Admin**

The Context Diagram for Admin illustrates how system administrators interact with PrenatalPlus through a dedicated admin interface. After secure login, the administrator accesses the User Management module, which allows the creation, modification, and maintenance of user accounts. Each user is assigned a unique identifier during registration to ensure proper tracking and access control.

The admin also has access to system monitoring features that provide an overview of user activity and system status. Through this interface, administrators can manage pregnancy data records, oversee reminder scheduling, and ensure smooth system operation. The backend infrastructure using Node.js and MongoDB supports real-time data updates and secure data handling, enabling effective system administration.

# 5. DETAILED DESIGN

## 5.1 Data Flow Diagram

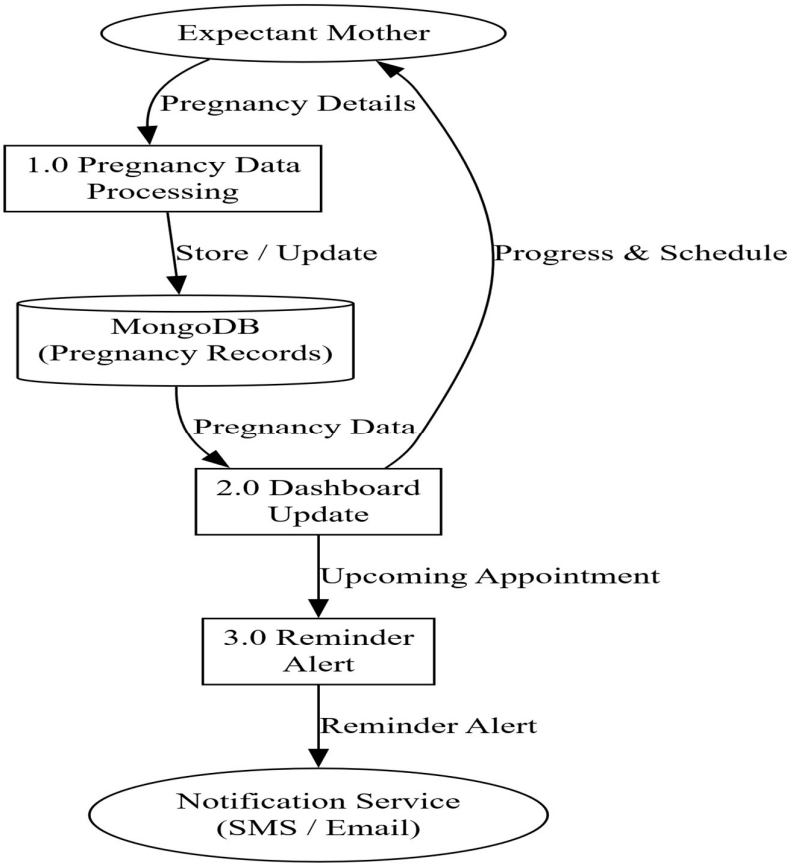


**Fig. 5.1: Dataflow diagram**

The Data Flow Diagram of the PrenatalPlus system can be described in the following points:

- Expectant mothers enter pregnancy-related details such as due date, trimester information, and appointment schedules into the system through the web interface.
- These inputs are transmitted securely to the backend server using RESTful APIs.
- The backend validates the received data and applies business rules related to pregnancy tracking and reminder scheduling.
- Validated pregnancy data and user profile information are stored in the MongoDB database for persistence and future reference.

- Simultaneously, appointment schedules and pregnancy progress data are sent to the dashboard module to update the user interface in real time.
- The system retrieves stored data whenever the user accesses the dashboard to display pregnancy progress, upcoming checkups, and health guidance.
- Reminder information flows to the notification module, which prepares alerts for upcoming prenatal appointments.
- Notification alerts are delivered to users through the configured notification service (SMS/Email), ensuring timely reminders.
- User actions such as updates to pregnancy details or appointments are continuously reflected back in the dashboard, maintaining data consistency.
- This closed-loop data flow ensures accurate prenatal monitoring and reliable healthcare support throughout the pregnancy period.

## 5.2 Activity Diagram

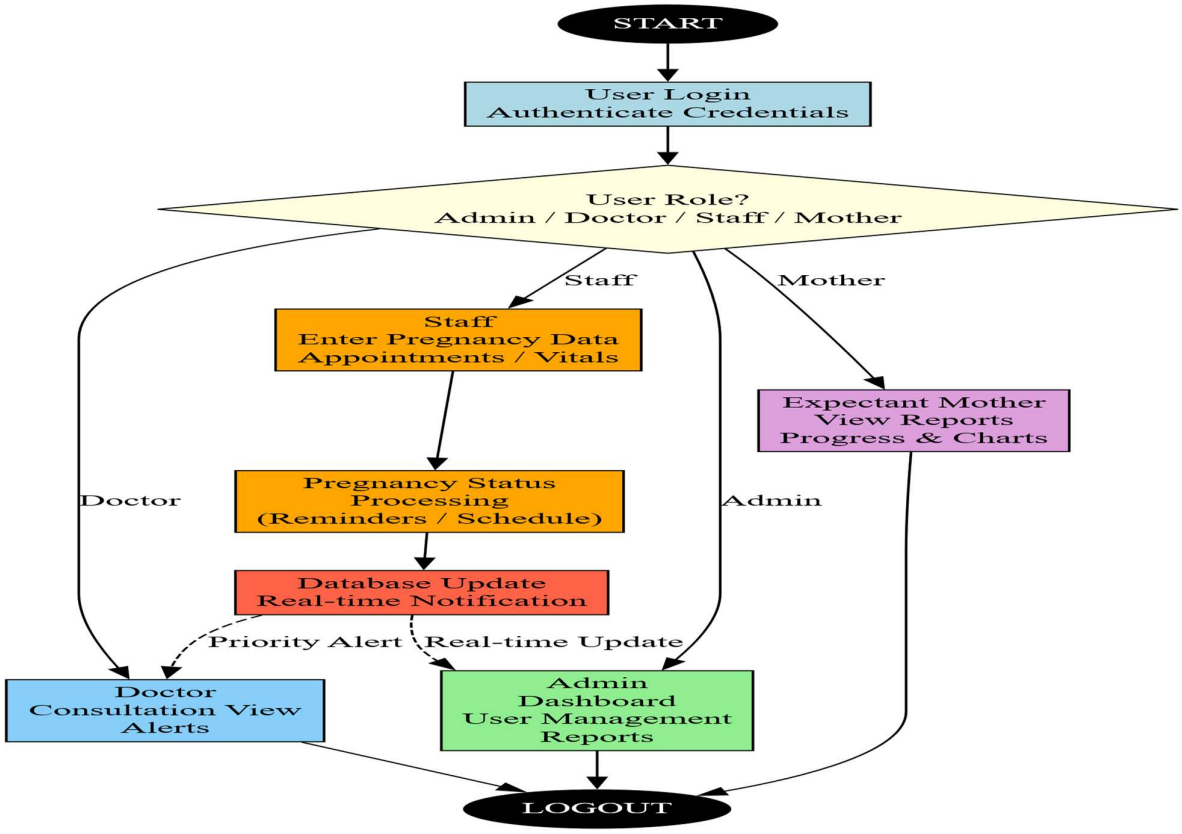


**Fig. 5.2 : Activity Diagram**

- The activity begins when the user accesses the PrenatalPlus system and submits login credentials.
- The authentication module validates the credentials and grants access upon successful verification.
- After authentication, the system identifies the user role (Expectant Mother or Admin) and directs the workflow accordingly.
- Expectant mothers proceed to view or update pregnancy profiles, check appointment schedules, or access prenatal health information.
- When pregnancy data is entered or modified, the system validates the input before saving it to the database.
- Appointment updates trigger the reminder scheduling process in the backend.
- The system updates the dashboard with the latest pregnancy progress and appointment information.
- Admin users follow a separate activity path that allows them to manage users, monitor system usage, and maintain data integrity.
- Error-handling activities manage invalid inputs or failed login attempts by redirecting users to the correction steps.
- Parallel background activities such as reminder checks and dashboard refreshes run alongside user actions.
- The activity concludes when the user logs out, terminating the session and returning the system to the initial state.
- Authorisation checks are enforced throughout the workflow to ensure users can access only permitted features.
- System validation checks ensure that mandatory pregnancy-related fields are completed before allowing data submission or updates.
- Session timeout mechanisms automatically log out inactive users to enhance system security and protect sensitive healthcare information.
- Audit logging activities record key user actions such as profile updates, appointment scheduling, and reminder triggers for monitoring and review purposes.
- The system performs periodic synchronization between frontend and backend to ensure that displayed data remains consistent and up to date.
- System notifications inform users about successful operations, warnings, or errors, improving usability and transparency during interactions.

## 5.3 Use Case Diagram

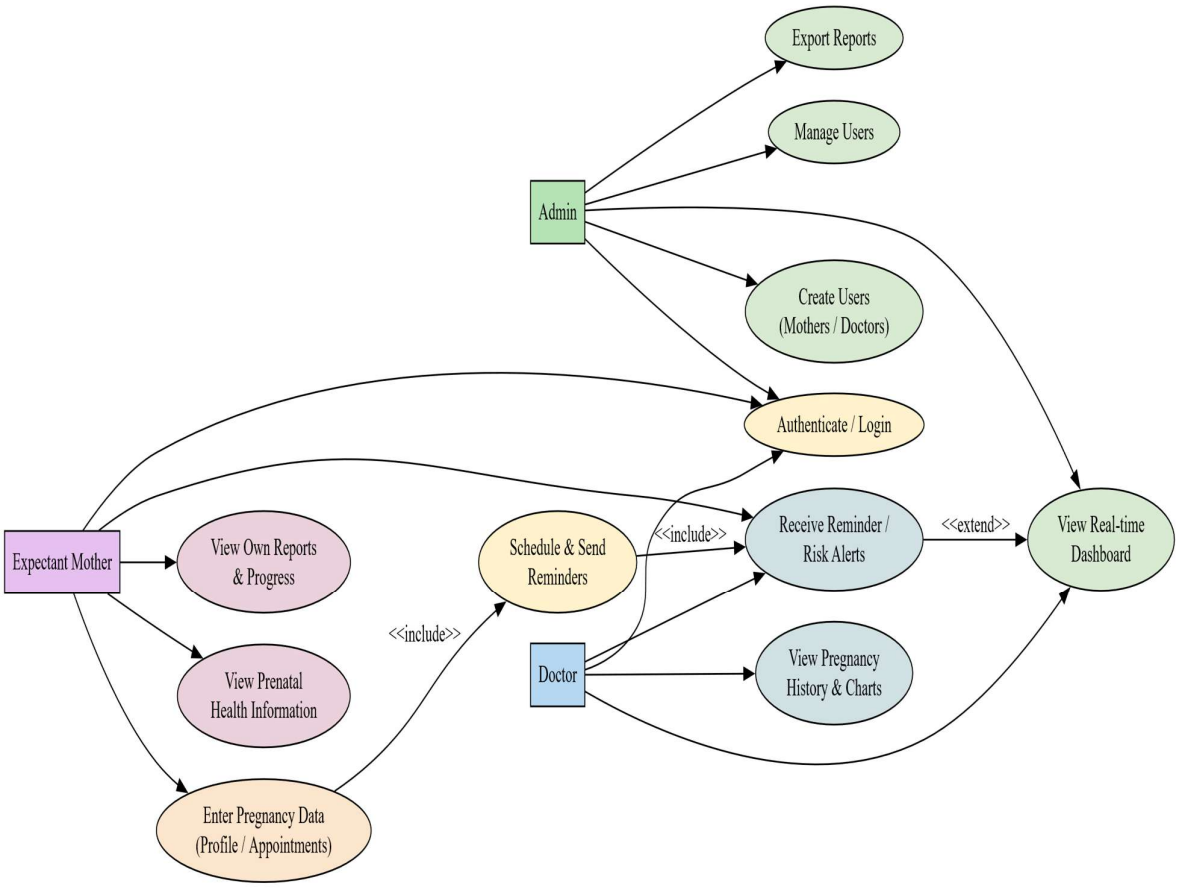


**Fig. 5.3 : Use Case Diagram**

The PrenatalPlus use case diagram depicts two primary actors: Expectant Mother and Administrator. Both actors interact with specific functions of the PrenatalPlus system.

All actors must use the Authenticate/Login use case to gain secure access to the system before performing any other actions. The Expectant Mother actor uses features such as Manage Pregnancy Profile, View Appointment Schedule, Receive Reminders, and Access Prenatal Health Information. These use cases allow users to track pregnancy progress and maintain regular prenatal care.

The Administrator actor accesses administrative use cases such as Create and Manage Users, Monitor System Activity, and Maintain Pregnancy Records. These use cases are restricted to admin users and are not accessible to expectant mothers.

The Schedule and Send Reminders use case is automatically triggered whenever appointment data is created or updated. The use case diagram visually represents the dependency between appointment management and reminder notification processes, highlighting how automated support is integrated into the system workflow.

### 5.4 Sequence Diagram

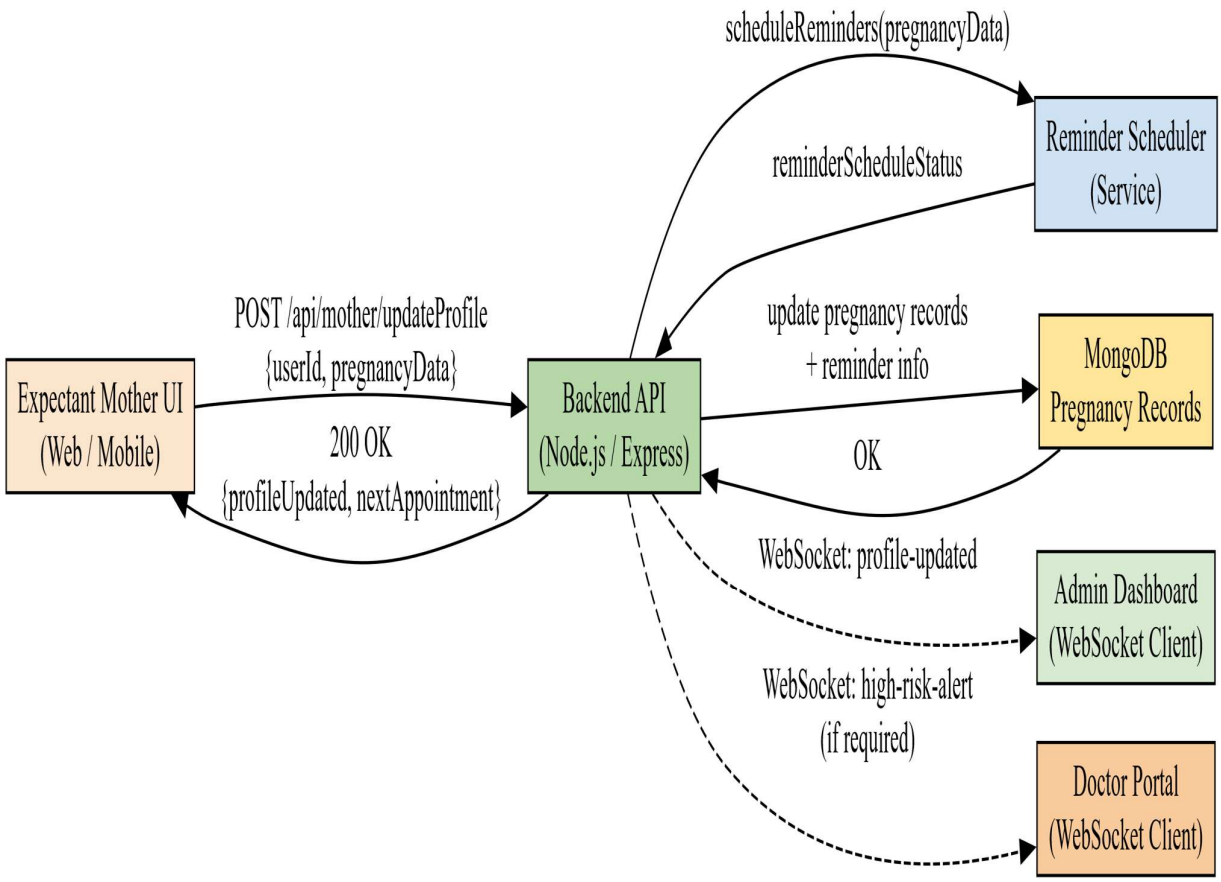


**Fig. 5.4 : Sequence Diagram**

The sequence diagram illustrates the interaction flow when an expectant mother updates pregnancy details in the PrenatalPlus system.

The sequence begins when the user submits updated pregnancy information through the React-based frontend interface. The frontend sends a POST request to the Backend API (Node.js/Express) containing user identification and pregnancy data. The Backend API validates the received data and processes the request.

Once validated, the backend updates the corresponding pregnancy records in the MongoDB database. After successful storage, the backend sends an HTTP success response back to the frontend. The frontend then refreshes the dashboard to display the updated pregnancy progress and appointment details.

If the update affects appointment schedules, the backend triggers the reminder scheduling logic. Notification services are prepared accordingly, ensuring future alerts are sent at the correct time. This sequence completes the request–response cycle while maintaining synchronized updates between the database, backend, and user interface.

# 6. IMPLEMENTATION

## Project Structure

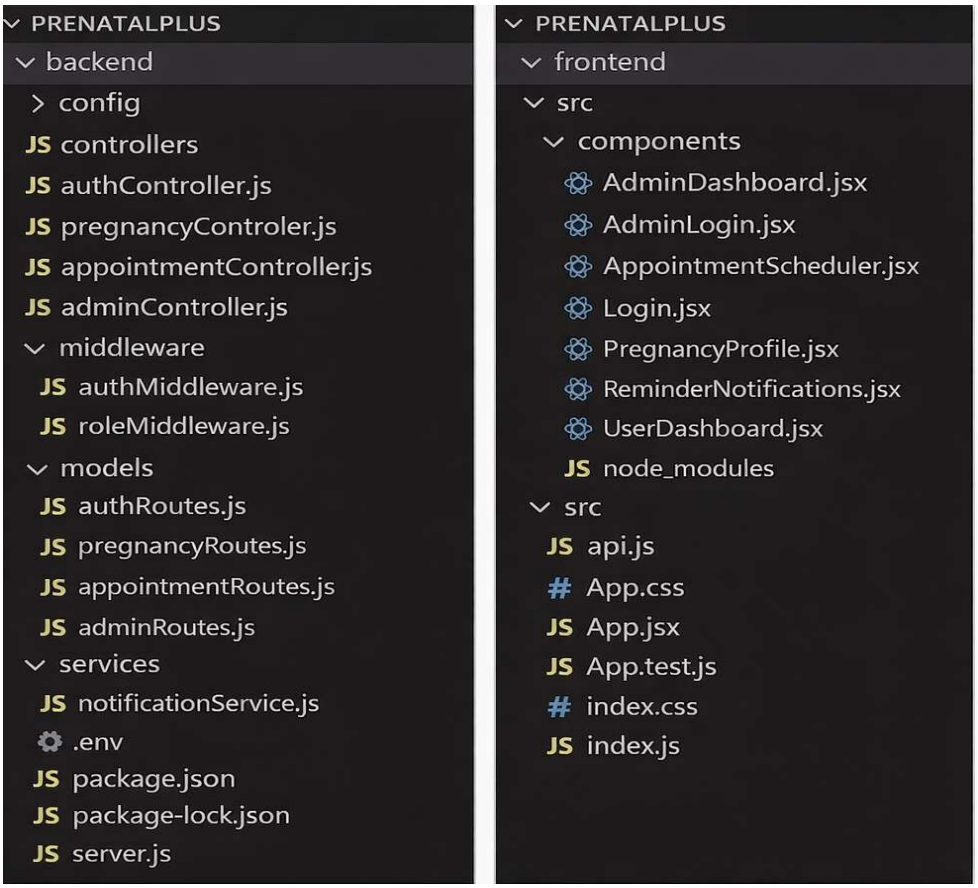


**Fig. 6.1: Project Structure**

## 6.1 Snippet code

### Backend Implementation

The backend of the PrenatalPlus system is designed to handle core application logic, secure data processing, and communication between the frontend and the database. It is developed using Node.js with the Express.js framework, which provides a lightweight and scalable server-side environment. The backend is responsible for managing user authentication, pregnancy data storage, appointment scheduling, and reminder notifications. MongoDB is used as the database to store user profiles and prenatal health records in a flexible document-oriented format. RESTful APIs are implemented to enable seamless interaction between the frontend interface and backend services, ensuring secure and efficient data exchange throughout the system.

### 6.1.1 server.js

This file is the entry point of the PrenatalPlus backend application. It initializes the server, establishes the database connection, configures middleware, and registers API routes.

```
// server.js

const express = require("express");

const dotenv = require("dotenv");

const cors = require("cors");

const connectDB = require("./config/db");

/* Load environment variables */

dotenv.config();

/* Initialize Express app */

const app = express();

/* Connect to MongoDB */

connectDB();

/* Middleware configuration */

app.use(cors());

app.use(express.json());

/* API Routes */

app.use("/api/auth", require("./routes/authRoutes"));

app.use("/api/pregnancy", require("./routes/pregnancyRoutes"));

app.use("/api/appointment", require("./routes/appointmentRoutes"));
```

```
app.use("/api/admin", require("./routes/adminRoutes"));

/* Default Route */

app.get("/", (req, res) => {

   res.send("PrenatalPlus Backend Server is Running");

});

/* Server Port */

const PORT = process.env.PORT || 5000;

/* Start Server */

app.listen(PORT, () => {

   console.log(`Server running on port ${PORT}`);

});
```

**6.1.2 Database Configuration (MongoDB)**

This file is responsible for establishing a secure connection between the backend server and the MongoDB database.

```
// config/db.js

const mongoose = require("mongoose");



const connectDB = async () => {

   try {

      await mongoose.connect(process.env.MONGO_URI, {

         useNewUrlParser: true,
```

```
        useUnifiedTopology: true
    });
    console.log("MongoDB Connected Successfully");
  } catch (error) {
    console.error("Database Connection Failed", error.message);
    process.exit(1);
  }
};
module.exports = connectDB;
```

**6.1.3 User Authentication Middleware**

This middleware ensures that only authenticated users can access protected routes.

```
// middleware/authMiddleware.js
const jwt = require("jsonwebtoken");
const protect = (req, res, next) => {
  const token = req.headers.authorization?.split(" ")[1];
  if (!token) {
    return res.status(401).json({ message: "Access Denied" });
  }
  try {
```

```
        const decoded = jwt.verify(token, process.env.JWT_SECRET);

        req.user = decoded;

        next();

    } catch (error) {

        res.status(401).json({ message: "Invalid Token" });

    }

};

module.exports = protect;
```

**6.1.4 Pregnancy Data Model**

This model stores pregnancy-related details for each registered user.

```
// models/Pregnancy.js

const mongoose = require("mongoose");

const pregnancySchema = new mongoose.Schema({

    userId: {

        type: mongoose.Schema.Types.ObjectId,

        ref: "User"

    },

    trimester: String,

    weight: Number,

    bloodPressure: String,
```

```
    lastCheckup: Date,

    riskLevel: String

}, { timestamps: true });

module.exports = mongoose.model("Pregnancy", pregnancySchema);
```

**6.1.5 Pregnancy Data Controller**

This controller handles pregnancy profile updates submitted by users.

```
// controllers/pregnancyController.js

const Pregnancy = require("../models/Pregnancy");

exports.updatePregnancyData = async (req, res) => {

    try {

        const data = await Pregnancy.create(req.body);

        res.status(201).json(data);

    } catch (error) {

        res.status(500).json({ message: "Failed to save pregnancy data" });

    }

};
```

## Frontend Implementation

The frontend of the PrenatalPlus system is developed using React.js, a component-based JavaScript library that enables the creation of interactive and responsive user interfaces. The frontend is responsible for handling user interactions, displaying pregnancy-related information, and communicating with the backend through REST APIs. Role-based components are designed for expectant mothers, doctors, and administrators to ensure a

personalized and secure user experience. The application follows a modular structure where each feature is implemented as a reusable React component, improving maintainability and scalability.

### 6.1.6 Main Application File (App.js)

The App.js file acts as the central component of the React application. It defines routing logic and connects different pages such as login, dashboards, and profile screens.

```
// src/App.js

import React from "react";

import { BrowserRouter as Router, Routes, Route } from "react-router-dom";

import Login from "./components/Login";

import PregnancyProfile from "./components/PregnancyProfile";

import UserDashboard from "./components/UserDashboard";

import AdminDashboard from "./components/AdminDashboard";

function App() {

  return (

    <Router>

      <Routes>

        <Route path="/" element={<Login />} />

        <Route path="/dashboard" element={<UserDashboard />} />

        <Route path="/profile" element={<PregnancyProfile />} />

        <Route path="/admin" element={<AdminDashboard />} />
```

```
        </Routes>
      </Router>
    );
}
export default App;
```

**6.1.7 API Integration Module**

This module is used to manage communication between the frontend and backend services.

```
// src/api.js
import axios from "axios";
const API = axios.create({
    baseURL: "http://localhost:5000/api"
});
export default API;
```

**6.1.8 Login Component**

This component handles secure user authentication.

```
// components/Login.jsx
import React, { useState } from "react";
const Login = () => {
    const [email, setEmail] = useState("");
    const [password, setPassword] = useState("");
```

```
  const handleLogin = (e) => {
    e.preventDefault();
    console.log("User Logged In");
  };
  return (
    <form onSubmit={handleLogin}>
      <input
        type="email"
        placeholder="Email"
        onChange={(e) => setEmail(e.target.value)}
      />
      <input
        type="password"
        placeholder="Password"
        onChange={(e) => setPassword(e.target.value)}
      />
      <button type="submit">Login</button>
    </form>
  );
};
```

```
export default Login;
```

**6.1.9 Pregnancy Profile Component**

This component allows expectant mothers to enter and update pregnancy-related information.

```
// components/PregnancyProfile.jsx

import React, { useState } from "react";

const PregnancyProfile = () => {

  const [trimester, setTrimester] = useState("");

  const [weight, setWeight] = useState("");

  const submitProfile = () => {

    alert("Pregnancy profile updated successfully");

  };

  return (

    <div>

      <input

        placeholder="Current Trimester"

        onChange={(e) => setTrimester(e.target.value)}

      />

      <input

        placeholder="Weight"

        onChange={(e) => setWeight(e.target.value)}
```

```
        />
        <button onClick={submitProfile}>Update Profile</button>
      </div>
    );
};
export default PregnancyProfile;
```

## 6.2 Screenshots

### Login Page

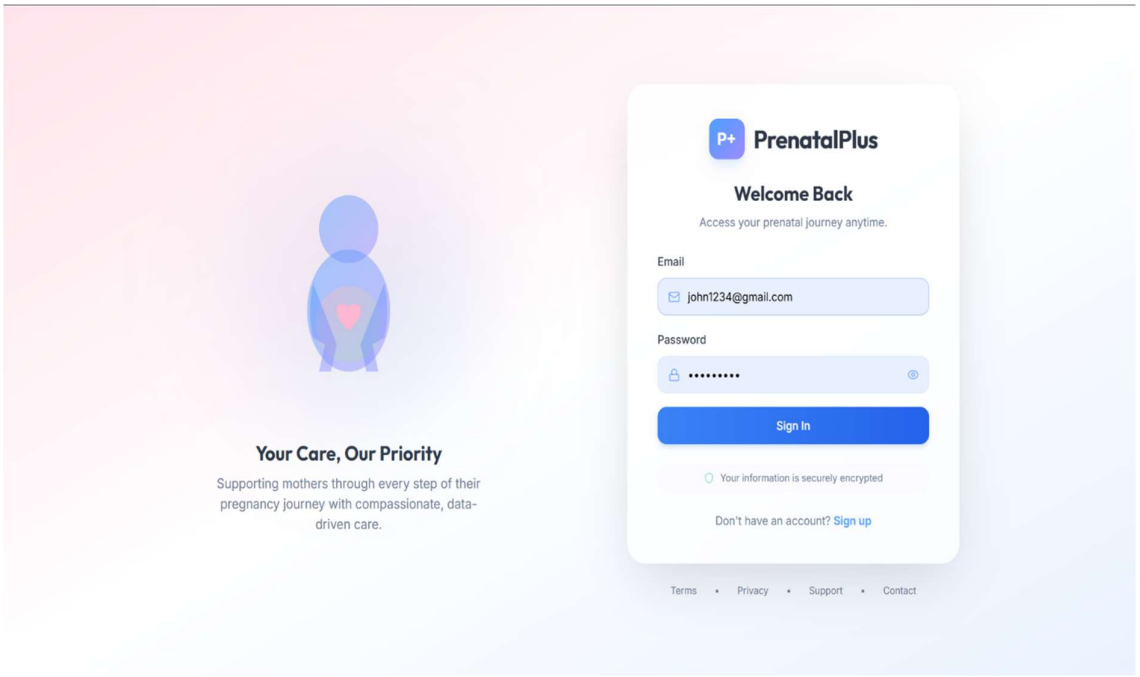


**Fig. 6.2 : Login Page**

## User Registration Page

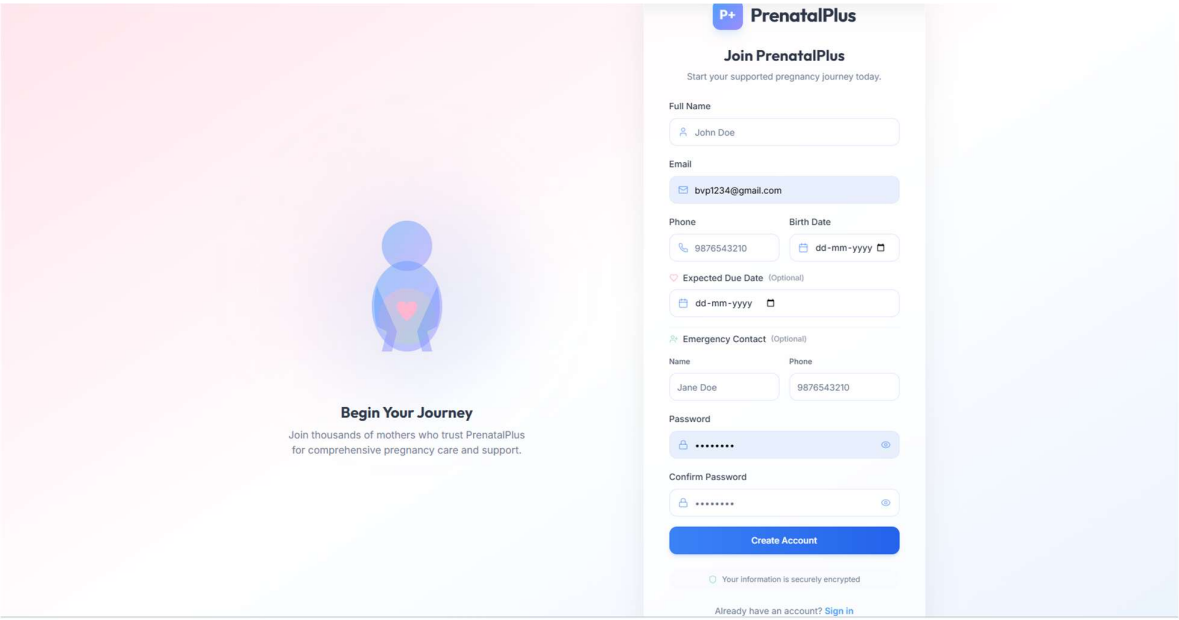


**Fig. 6.3 : User Registration Page**

## User Dashboard

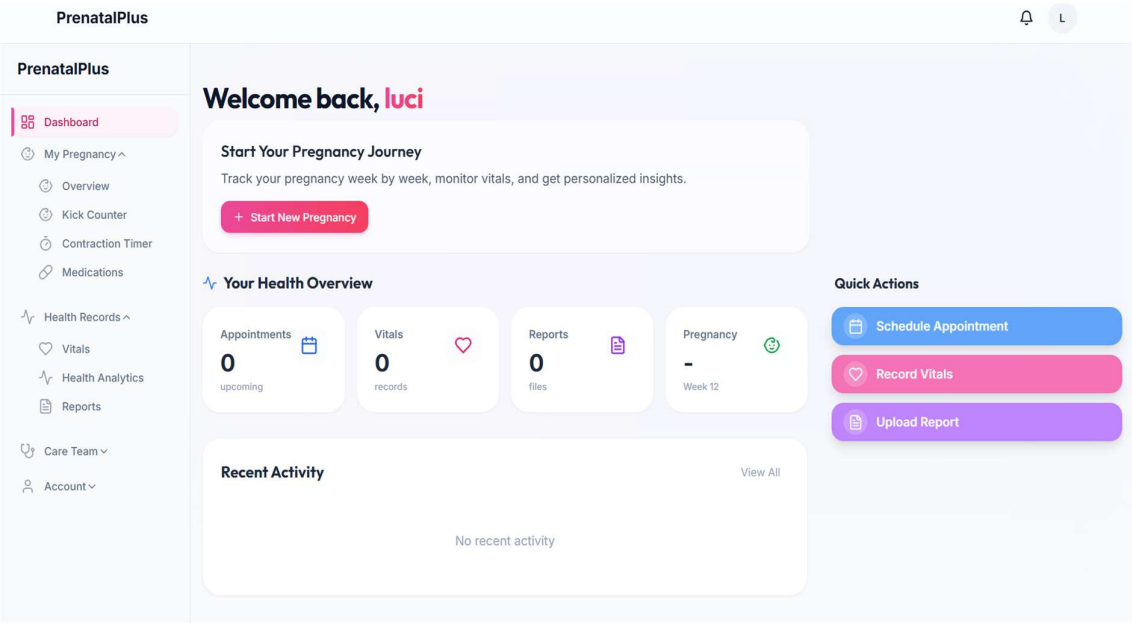


**Fig. 6.4 : User Dashboard**

## Pregnancy Profile Management Page

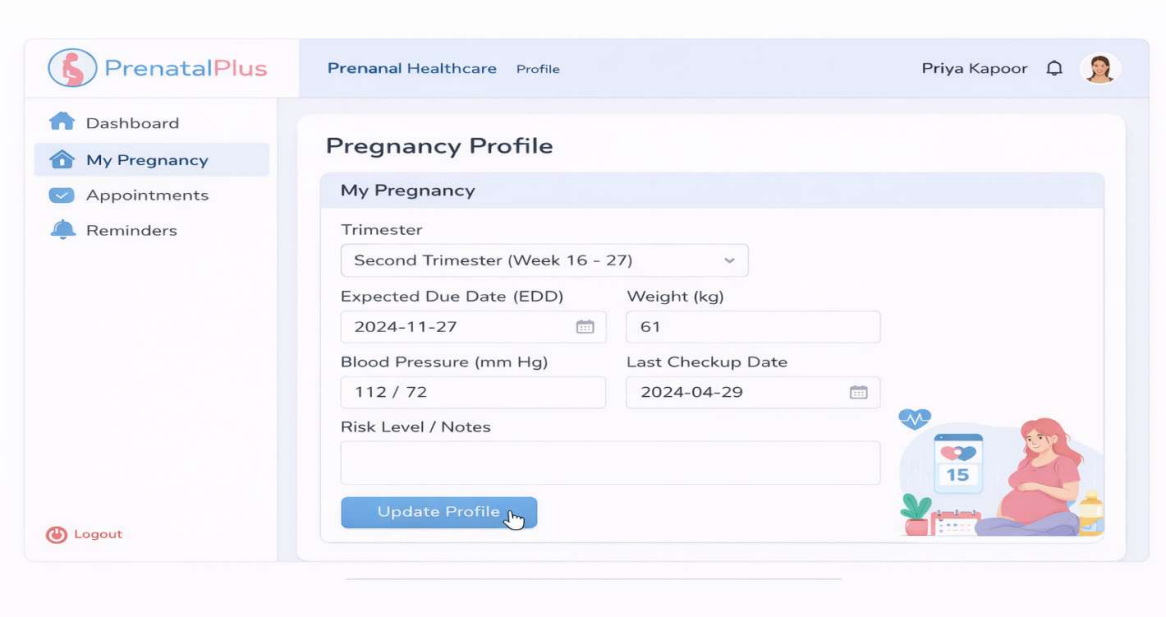


**Fig. 6.5 : Pregnancy Profile Management Page**

## Appointment Scheduling Page

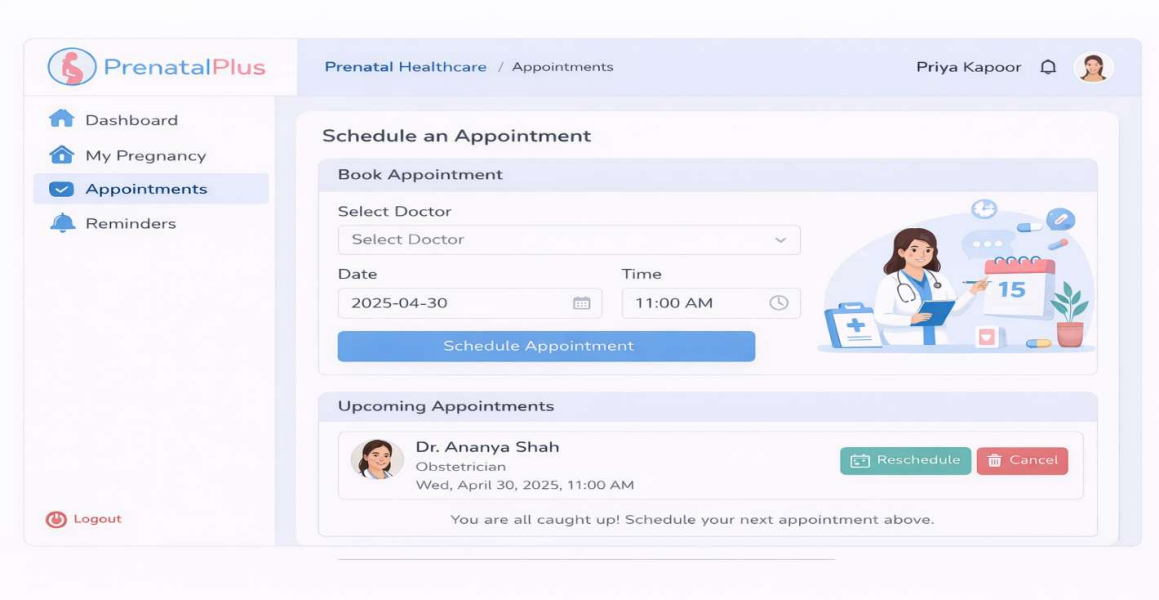


**Fig. 6.6 : Appointment Scheduling Page**

## Reminder Notification Display

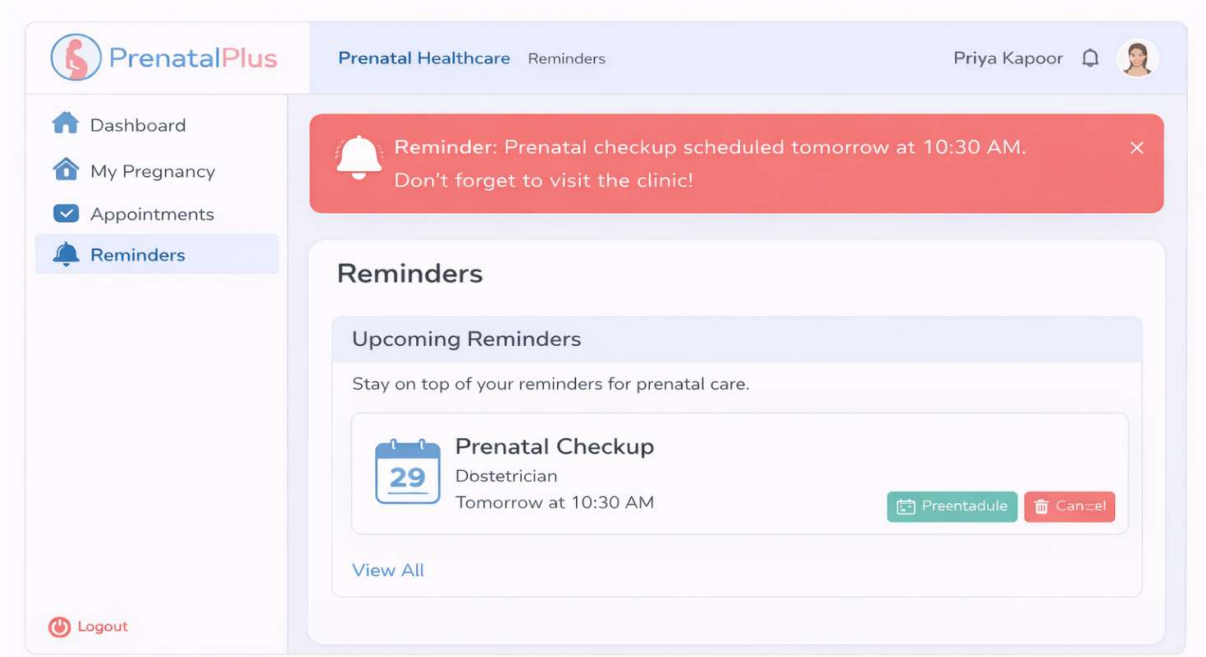


**Fig. 6.7 : Reminder Notification Display**

## Admin Login Page

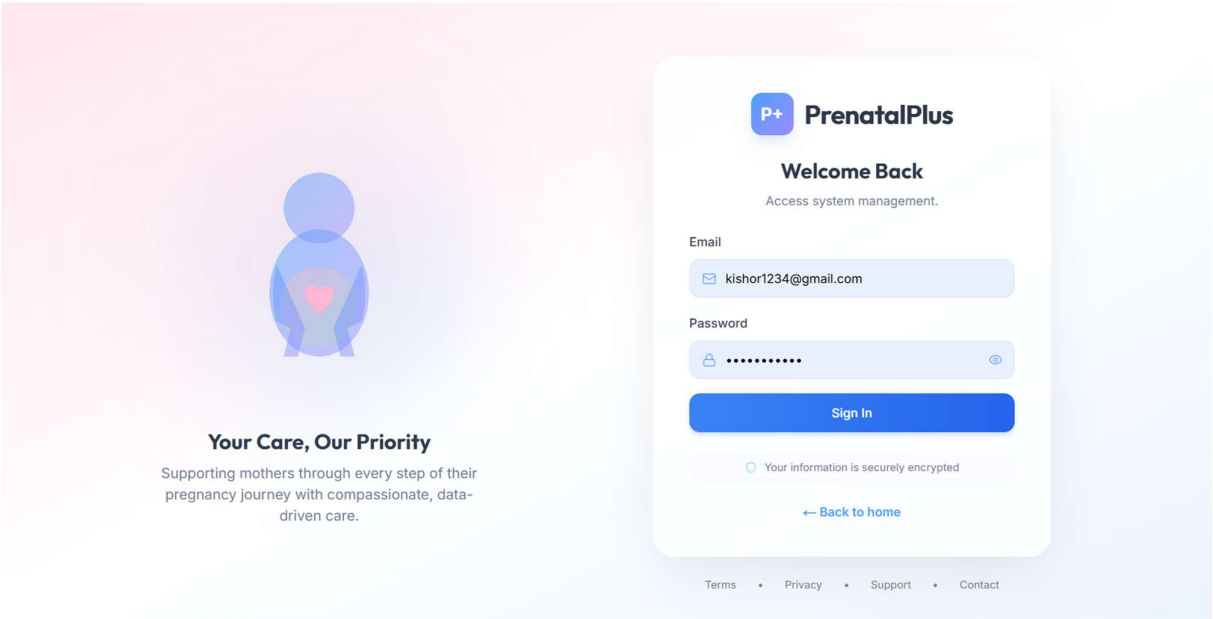


**Fig. 6.8 : Admin Login Page**

## Admin Dashboard

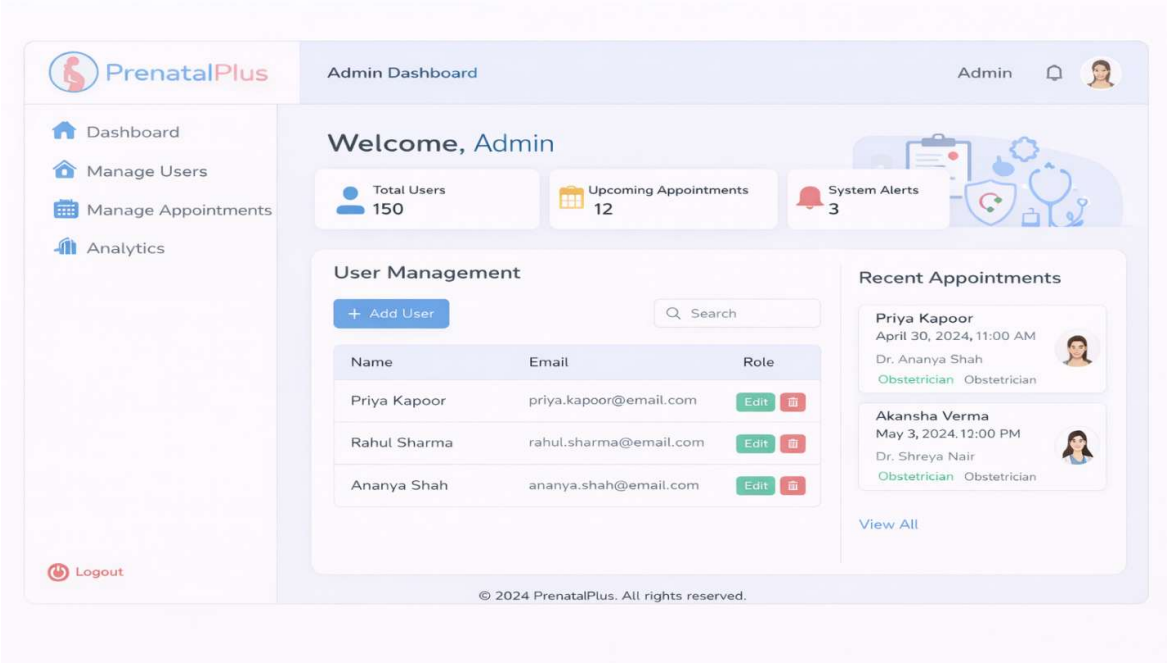


**Fig. 6.9 : Admin Dashboard**

## User Management Module

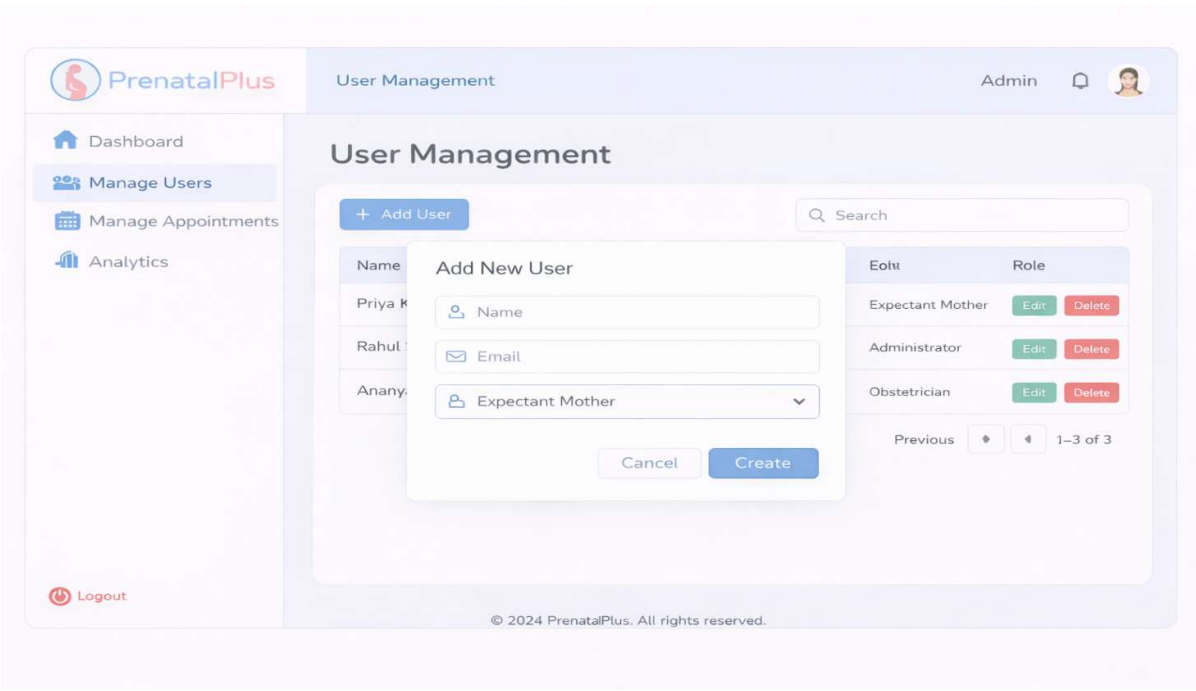


**Fig. 6.10 : User Management Module**

# 7. SOFTWARE TESTING

The software testing of the PrenatalPlus system focuses on validating the correctness of pregnancy data handling, the reliability of reminder notifications, and the enforcement of secure role-based access. Testing ensures that the system performs as expected under different user scenarios and that sensitive prenatal information is handled safely and accurately.

## 7.1 Unit Testing

Unit testing focuses on testing individual modules and functions in isolation to verify that each component performs its intended task correctly.

**Backend Unit Testing**

- The pregnancy data processing functions, such as updatePregnancyData() and getPregnancyProfile(), are tested to ensure correct storage and retrieval of user-entered information.
- Authentication-related methods, including user login and token validation, are tested to verify secure access control.
- Appointment scheduling and reminder generation functions are tested to ensure reminders are created correctly based on appointment dates.
- Controller methods such as user registration, profile update, and appointment retrieval are tested using testing tools like Jest or Mocha.
- Database interaction functions are tested to confirm successful read and write operations in MongoDB.

**Frontend Unit Testing**

- Individual React components, such as Login, UserDashboard, PregnancyProfile, and AdminDashboard, are tested separately to ensure proper rendering and logic execution.
- Form validation logic is tested to verify correct handling of user input and error messages.
- API helper functions used for communicating with the backend are tested to ensure correct request formatting and response handling.

- Component state updates and navigation flows are tested to validate smooth user interaction.

## 7.2 Automation Testing

Automation testing is performed to validate end-to-end user workflows across different roles in the PrenatalPlus system.

Typical Automated Test Scenarios

- **Admin Flow:**
  Login → access admin dashboard → create or manage user accounts → monitor system usage and data records.
- **Expectant Mother Flow:**
  Login → update pregnancy profile → view pregnancy progress → schedule or view appointments → receive reminder notification.
- **Reminder Flow:**
  Appointment creation or update → system schedules a reminder → reminder notification is triggered and delivered at the correct time.
- **System Validation Flow:**
  Invalid login credentials → error message displayed → user redirected to retry login. Missing mandatory pregnancy fields → validation error shown → submission blocked until corrected.

These automated tests ensure that all major system workflows function correctly and consistently across different user roles.

## 7.3 Sample Test Cases

**Table 7.1: Test Cases**

| TC ID | Test Case Name | Precondition | Steps | Expected Result |
|---|---|---|---|---|
| TC01 | Admin Login (Positive) | Admin credentials (admin/xxx) are configured in the | 1. Open the application. 2. Click on “Admin Login. 3. Enter valid | Admin is redirected to the Admin Dashboard. API /api/admin/login returns status success: true. |

| | | | | |
|---|---|---|---|---|
| | | admin authentication module. | admin credentials. 4. Click “Submit. | Dashboard displays user and pregnancy records if available. |
| TC02 | Expectant Mother Login | A registered expectant mother account exists in the system. | 1. Open the application. 2. Click on “User Login. 3. Enter a valid email and password. 4. Click “Login. | User is redirected to the User Dashboard. API returns an authentication token. Pregnancy details are loaded successfully. |
| TC03 | Add / Update Pregnancy Profile | User is logged in, and a pregnancy profile exists. | 1. Log in as an expectant mother. 2. Open Pregnancy Profile page. 3. Enter trimester, weight, and health details. 4. Click “Update Profile. | Pregnancy data is saved in MongoDB. API /api/pregnancy/update returns a success response. Updated details are reflected on the dashboard. |
| TC04 | Appointment Scheduling | User is logged in, and the pregnancy profile is active. | 1. Log in as an expectant mother. 2. Open the Appointment section. 3. Select date and time. 4. Click “Schedule Appointment. | The appointment record is stored in the database. Confirmation message is displayed. The reminder schedule is generated successfully. |
| TC05 | Reminder Notification Trigger | An appointment exists with an upcoming date. | 1. System checks scheduled appointments. 2. Reminder time threshold is reached. | A reminder notification is generated. The notification service sends SMS/Email alerts. User receives a reminder message. |
| TC06 | View Pregnancy Reports | User has at least one pregnancy record stored. | 1. Log in as an expectant mother. 2. Open “My Reports. 3. Select the latest record. | Pregnancy details and progress are displayed correctly. No other user’s data is visible. |
| TC07 | Admin User Management | Admin is logged in successfully. | 1. Open Admin Dashboard. 2. Navigate to User Management. 3. Add or update user details. | User data is updated in the database. Changes are reflected immediately in the admin panel. |
| TC08 | Security: Unauthorised Access | User is not authenticated, or the token is missing. | 1. Access the protected API endpoint directly. 2. Do not include the authentication token. | API responds with HTTP 401 Unauthorised. Access is denied, and no data is exposed. |

# 8. CONCLUSION

PrenatalPlus Web Application offers an efficient, user-friendly, and reliable digital solution for prenatal healthcare management. The system eliminates manual record keeping and automates pregnancy tracking, appointment scheduling, and reminder notifications to provide expectant mothers with continuous and organised prenatal support. The application delivers a structured digital workflow that assists users in monitoring pregnancy progress and accessing essential healthcare information promptly.

The application is developed using MERN full-stack technology, with a React-based frontend and a Node.js/Express.js backend integrated with MongoDB for secure data storage. This architecture ensures scalability, responsiveness, and reliable data handling. PrenatalPlus successfully manages pregnancy-related data, user profiles, and appointment records through RESTful APIs, enabling smooth interaction between system components.

## What was accomplished?

- Expectant mothers are provided with a centralised platform to enter and update pregnancy details, view progress reports, and manage prenatal appointments, eliminating the need for paper-based tracking and reducing dependency on manual reminders.
- Automated reminder functionality ensures that users receive timely notifications for scheduled prenatal checkups and important health-related activities, improving adherence to recommended medical visits.
- Admin users are equipped with management tools to oversee user accounts, monitor system usage, and maintain data integrity, ensuring smooth system operation and controlled access.
- The role-based system design ensures data privacy, allowing users to access only their own records while administrators manage system-level operations securely.
- The responsive user interface improves accessibility across devices, enabling consistent use of the application throughout the pregnancy period.

## Performance Validation

System testing and validation results demonstrate that the PrenatalPlus application performs reliably under expected usage conditions. User authentication, pregnancy data storage, appointment scheduling, and reminder delivery were validated successfully through unit and automation testing. The system exhibited stable performance, secure access control, and consistent data handling, indicating readiness for practical deployment.

## Future Scope

Phase I confirms that PrenatalPlus can support a scalable infrastructure suitable for broader adoption. The system architecture allows for seamless integration of additional features in future phases. Potential enhancements include integration with wearable health devices for real-time vitals monitoring, AI-based pregnancy risk assessment, multilingual support, and mobile application deployment. With its modular and extensible design, PrenatalPlus is well-positioned for future advancements in digital prenatal healthcare.

# 9. SCOPE FOR PHASE-II

## MOBILE APPLICATION DEVELOPMENT FOR PRENATALPLUS

**Mobile Platform Deployment:**

- **Target Platforms:**
  Native/cross-platform Android and iOS mobile application developed using React Native or Flutter to extend PrenatalPlus beyond the web.

- **User Coverage:**
  Supports expectant mothers, administrators, and healthcare staff with role-based access. Designed to scale from individual users to large healthcare institutions.

- **Device Compatibility:**
  Optimised for smartphones and tablets with Android 10+ and iOS 14+ versions, ensuring smooth performance across devices.

**Enhanced Mobile Features:**

- **Mobile-First Access:**
  On-the-go access to pregnancy tracking, appointment schedules, and health information through a user-friendly mobile interface.
- **Push Notifications:**
  Instant push notifications for appointment reminders, medication alerts, and important prenatal updates.
- **Offline Support:**
  View previously synced pregnancy data in offline mode with automatic synchronisation once internet connectivity is restored.
- **Secure Biometric Login:**
  Support for fingerprint or face authentication to enhance security and ease of access.

**Backend Extensions (Minimal Changes Required):**

- /api/mobile/auth – Optimised authentication endpoints for mobile clients.
- /api/mobile/profile – Lightweight APIs for pregnancy data retrieval and updates.
- /api/mobile/notifications – Push notification service integration (Firebase / Apple Push).

The existing backend APIs can be reused with minimal or no refactoring, ensuring faster Phase-II implementation.

## 3-Month Timeline (Total: 12 Weeks)

**Table 9.1: Phase-II Mobile App Development Timeline**

| Milestone | Weeks | Deliverables |
|---|---|---|
| Month 1 | 1–4 | Mobile UI design, navigation flow, API integration planning |
| Month 2 | 5–8 | Core feature development, push notifications, and offline support |
| Month 3 | 9–12 | Testing, optimisation, deployment, training & go-live |

# 10. BIBLIOGRAPHY


- World Health Organisation. WHO Recommendations on Antenatal Care for a Positive Pregnancy Experience. World Health Organisation, Geneva, 2023.
- American College of Obstetricians and Gynaecologists. Guidelines for Prenatal Care. ACOG Practice Bulletin, 2022.
- Kumar, S., and Verma, R. "Digital Health Applications for Prenatal Monitoring in Resource-Constrained Settings." International Journal of Health Informatics, vol. 15, no. 2, 2024, pp. 85–102.
- Albahri, O. S., et al. "Real-Time Remote Health Monitoring Systems: A Systematic Review." Journal of Medical Systems, vol. 43, no. 6, 2019, pp. 1–18.
- MongoDB Inc. MongoDB Documentation: Data Modelling and Performance Best Practices. 2023. Available at: https://www.mongodb.com/docs/
- Facebook Open Source. React: A JavaScript Library for Building User Interfaces. 2023. Available at: https://reactjs.org/
- Node.js Foundation. Node.js Official Documentation. 2023. Available at: https://nodejs.org/
- Express.js. Express – Fast, Unopinionated, Minimalist Web Framework for Node.js. 2023. Available at: https://expressjs.com/
- Twilio Inc. Patient Engagement and Reminder Systems Using Twilio Communication APIs. 2023. Available at: https://www.twilio.com/healthcare
- Patel, M., and Shah, N. "Secure Web-Based Healthcare Applications Using MERN Stack." Journal of Computer Applications in Healthcare, vol. 11, no. 1, 2024, pp. 33–48.

# 11. APPENDIX A: USER MANUAL

## System Overview

PrenatalPlus is a web-based prenatal healthcare management system designed to support expectant mothers by digitally managing pregnancy-related information. The system allows users to register, maintain pregnancy profiles, schedule prenatal appointments, and receive automated reminders. Administrators manage users and monitor system activity, while the application ensures secure, role-based access to sensitive healthcare data.

The system replaces manual pregnancy tracking with a structured digital workflow, providing continuous access to pregnancy progress and reminders to improve adherence to prenatal care.

## Required Components

- **Server / Backend:**
  Node.js with the Express.js framework, MongoDB database for data storage.
- **Frontend:**
  React.js for user interface development, a modern web browser (Google Chrome, Microsoft Edge, or Mozilla Firefox).
- **Utilities:**
  Visual Studio Code (or equivalent IDE), Git for version control, Postman for API testing, MongoDB Compass (optional), environment configuration file (.env) containing database connection string and security keys.

## Basic Setup and Configuration

- Install Node.js and MongoDB on the development or deployment machine.
- Clone the PrenatalPlus project repository and install backend and frontend dependencies using a package manager (npm install).
- Configure the .env file with MongoDB connection URI, server port, and security-related environment variables.
- Start the backend server using Node.js and ensure database connectivity.
- Start the frontend React application and verify that the login page is accessible through a web browser.

- Confirm that API communication between frontend and backend is functioning correctly.

## How to Operate the System

### Admin

1. Log in using admin credentials and access the Admin Dashboard.
2. Create and manage user accounts for expectant mothers and system administrators.
3. Monitor system usage and manage stored pregnancy and appointment data.
4. Ensure proper access control and system configuration.

### Expectant Mother

1. Log in using registered credentials.
2. Access the user dashboard to view pregnancy progress and upcoming appointments.
3. Enter or update pregnancy-related information such as trimester details and health notes.
4. View scheduled appointments and receive reminder notifications.

### Safety and Usage Guidelines

- The system is intended to support prenatal care tracking and should not replace professional medical consultation.
- Users should ensure that login credentials are kept confidential to prevent unauthorised access.
- Sensitive healthcare data should be accessed only through secure and trusted networks.
- Regular database backups should be maintained to prevent data loss.

### Troubleshooting Guide

### Issue 1: Application Not Loading or Backend Unreachable

- **Possible Causes:** Backend server not running, incorrect API configuration, or database connection issues.

- **Solutions:** Verify that the backend server and MongoDB are running, check environment variables, and inspect server logs for errors.

**Issue 2: Unable to Save or Update Pregnancy Data**

- **Possible Causes:** Missing required fields, invalid input format, or API failure.
- **Solutions:** Ensure all mandatory fields are filled correctly and verify backend API connectivity.

**Issue 3: Reminder Notifications Not Triggered**

- **Possible Causes:** Appointment not scheduled properly or reminder service not active.
- **Solutions:** Verify appointment dates, check backend reminder logic, and inspect system logs.

**Issue 4: Login Failures for Valid Users**

- **Possible Causes:** Incorrect credentials, role mismatch, or user record not found.
- **Solutions:** Recheck login details, ensure correct role selection, and confirm user existence in the database.

**Issue 5: Data Not Updating on Dashboard**

- **Possible Causes:** Network latency or API response delay.
- **Solutions:** Refresh the page, verify internet connectivity, and check backend service status.

## Conclusion

This user manual provides step-by-step guidance for operating the PrenatalPlus system safely and effectively. Following these instructions ensures smooth system usage, secure data handling, and reliable prenatal healthcare support.

# 12. APPENDIX B: SDG COMPLIANCE

## i. Project Title:

PrenatalPlus – Prenatal Healthcare Management System

## ii. Relevant Sustainable Development Goals (SDGs):

**Table 12.1: Mapping of SDG Goals**

| SDG No. | Name | Relevance to Project |
|---|---|---|
| SDG 3 | Good Health and Well-Being | Supports maternal health by enabling regular pregnancy monitoring, appointment reminders, and access to prenatal healthcare information |
| SDG 9 | Industry, Innovation and Infrastructure | Utilises digital healthcare technology and web-based infrastructure to improve accessibility and efficiency of prenatal care |

## iii. SDG Mapping Justification

- Supports SDG 3 by promoting safe motherhood through timely prenatal checkups, improved awareness, and structured pregnancy tracking, which can help reduce maternal and neonatal health risks.
- Advances SDG 9 by leveraging modern web technologies to deliver innovative digital healthcare solutions that enhance healthcare service delivery and system efficiency.

## iv. Interdisciplinary / Multidisciplinary Aspects

**Table 12.2: Inter / Multidisciplinary Aspects Involved in the Project**

| Discipline | Contribution |
|---|---|
| Computer Science | Designed and developed the web application using MERN stack technologies |
| Healthcare / Medical Science | Provided domain knowledge related to prenatal care guidelines and maternal health practices |

| | |
|---|---|
| Information Technology | Supported system deployment, data management, and secure application infrastructure |

## v. Societal / Environmental Impact

- Target Beneficiaries: Expectant mothers, healthcare providers, and healthcare administrators.
- Positive Outcomes: Improved prenatal care adherence, better maternal health awareness, reduced missed appointments, and enhanced access to digital healthcare services.
- Environmental Impact: Reduction in paper-based records and manual documentation contributes to environmentally sustainable healthcare practices.

# 13. APPENDIX C: WORK BREAKDOWN STRUCTURE

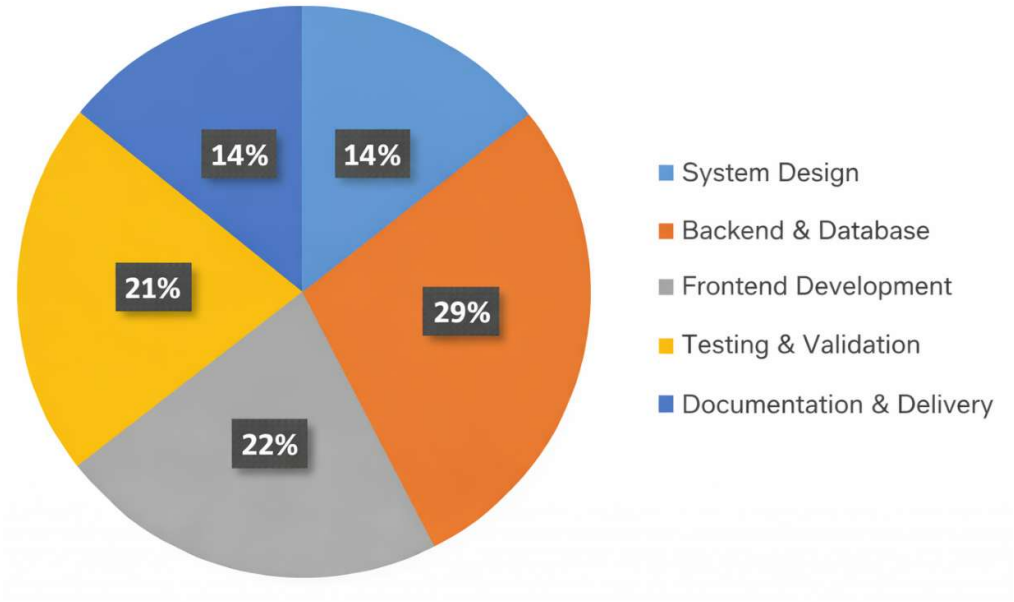


**Fig. 13.1: SDLC Work Breakdown Structure**

## Phase 1: System Design

- **Requirement Analysis:**
  Gather requirements related to prenatal healthcare management, pregnancy tracking, appointment scheduling, reminder notifications, and user roles. Identify workflows for expectant mothers and administrators, along with data requirements.
- **System Architecture Design:**
  Design the three-tier architecture using React.js, Node.js, and MongoDB. Prepare system architecture, context diagrams, and data flow diagrams.
- **Data Model Design:**
  Define database schemas for users, pregnancy profiles, and appointments. Identify data fields, relationships, and validation rules.
- **Workflow and UI Planning:**
  Map user workflows for expectant mothers and admins. Prepare wireframes and UI flow diagrams for dashboards and profile management screens.

## Phase 2: Backend & Database Development

- **Environment Setup:**
  Configure Node.js, Express.js, MongoDB, project structure, and environment variables.
- **API and Authentication Development:**
  Develop REST APIs for user registration, login, pregnancy data management, and appointment handling. Implement JWT-based authentication and role-based access control.
- **Database Implementation:**
  Create MongoDB collections and implement data storage, retrieval, and update logic.
- **Reminder and Notification Services:**
  Implement backend logic for scheduling reminders and triggering notification alerts.

## Phase 3: Frontend Development

- **User Interface Development:**
  Develop React-based components for login, dashboard, pregnancy profile, and appointment management.
- **Admin Portal:**
  Implement an admin dashboard for user management and system monitoring.
- **API Integration:**
  Integrate frontend components with backend APIs for data exchange and real-time updates.

## Phase 4: Testing and Validation

- **Unit and Integration Testing:**
  Test backend APIs, database operations, and frontend components individually and collectively.
- **Functional and Performance Testing:**
  Validate end-to-end workflows such as login, pregnancy profile updates, appointment scheduling, and reminders.

- **Security and Usability Review:**
  Test authentication, authorisation, and data privacy mechanisms. Review usability through user feedback.

## Phase 5: Documentation and Delivery

- **Technical Documentation:**
  Prepare system architecture diagrams, database schema descriptions, and API documentation.
- **User Manual Preparation:**
  Develop installation and usage guides for the admin and expectant mother roles.
- **Final Report and Presentation:**
  Compile the academic project report and prepare demonstration materials for submission.

**Effort Distribution**

- System Design – **15%**
- Backend & Database – **30%**
- Frontend Development – **25%**
- Testing & Validation – **15%**
- Documentation & Delivery – **15%**

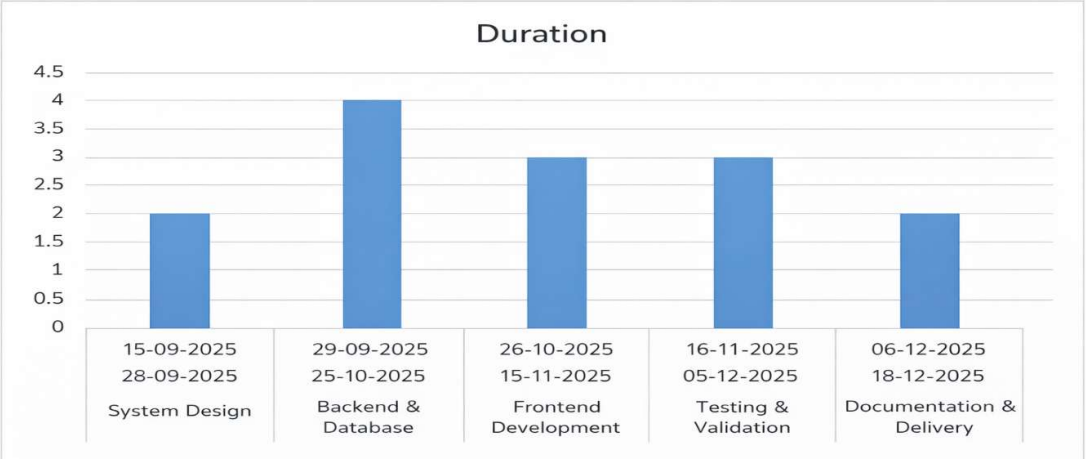


**Fig. 13.2 : Project Phase Timeline**

**Table 13.1: Project Timeline**

| Phase | Start Date | End Date | Duration |
|---|---|---|---|
| System Design | 15-09-2025 | 28-09-2025 | 2 Weeks |
| Backend & Database | 29-09-2025 | 25-10-2025 | 4 Weeks |
| Frontend Development | 26-10-2025 | 15-11-2025 | 3 Weeks |
| Testing & Validation | 16-11-2025 | 05-12-2025 | 3 Weeks |
| Documentation & Delivery | 06-12-2025 | 18-12-2025 | 2 Weeks |

# 14. APPENDIX D: PLAGIARISM REPORT